Why Do Retirees Retain Wealth? Quasi-Experimental Evidence from the Netherlands*

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Abstract

I study why retired households draw down so little wealth, a puzzling empirical pattern known as the *retirement savings puzzle*. Using population-wide administrative data from the Netherlands, I investigate the role of long-term-care expenditure risk and bequest motives in explaining this pattern. For identification, I exploit statutory kinks in the Dutch policy rules governing costs of care, and examine life events that change the set of potential heirs. I find both motives are important drivers of old-age wealth. A marginal euro increase in care costs increases annual wealth accumulation by 45 cents, implying a precautionary stock of up to 22 percent of wealth. On the bequest side, losing an adult child decreases wealth by 5.5 percent, while gaining an unexpected grandchild increases it by 1.2 percent five years after the event. Findings provide evidence of an *intentional* bequest motive directed toward particular heirs. Together, the two motives account for up to 40 percent of old-age wealth, but leave a substantial portion unexplained.

Keywords: retirement, savings, aging, long term care, bequests, insurance, wealth
JEL codes: D14, G51, H31

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1 Introduction

Population aging puts the economic behavior of old age households at the core of public policy. Consumption and saving decisions of retirees can inform the design of social security, health and long term care systems, and taxation of intergenerational transfers. Standard life-cycle models predict households save during their working lives and draw down their savings in retirement. However, empirical studies find little dissaving in old age (Mirer, 1979; Poterba et al., 2011), an empirical regularity known as the *Retirement Savings Puzzle* (French et al., 2023). This gap between the predicted and observed wealth drawdown points to the existence of motives for holding wealth late in life.

There are two prevailing motives which raise old-age saving: precautionary savings against health-related costs, and bequests. It has been difficult to empirically disentangle these two motives, because they produce nearly observationally equivalent behavior. Households concerned about future health-related costs save more today as a form of self-insurance. Bequest-motivated households have similar saving patterns, as they reserve a portion of their wealth to be transferred to heirs. Despite considerable effort to quantify these motives, their importance in explaining the puzzle remains an open question, as wealth profiles fit both explanations similarly well (De Nardi et al., 2016; French et al., 2023).

In this paper, I investigate the extent to which precautionary and bequest motives individually contribute to wealth held in old age. Using high-quality administrative data from the Netherlands, I find that both long-term-care (LTC) expenditure risk and bequest motives drive savings up substantially. Additionally, results point to an *intentional* bequest motive directed towards particular beneficiaries, rather than purely accidental bequests generated by mortality risk or incidental bequests left over from precautionary saving. However, even though these two motives have a significant impact on wealth at old age, they can only account for up to 40% of wealth held, leaving a large portion unexplained.

Quantifying the effects of precautionary and bequest motives is challenging for two primary reasons. First, it is difficult to find exogenous variation in expected care costs and bequest motives that allows for causal identification of these effects. In many settings, care costs depend on endogenous decisions of the household, and bequest preferences are unobserved. Second, even with such variation, researchers need detailed data combining information on wealth, income, health-related expenditures and family composition.

The Netherlands provides an especially useful setting for studying these motives, combining a generous welfare state, quasi-experimental policy variation, and unusually rich administrative data. To start with, the Dutch welfare state already insures households against a variety of risks old-age households face. For instance, the universally covered health insurance limits the yearly out-of-pocket spending to €635, and the pension system consists of a mandatory occupational pension and state pension that together make up a 75% mean replacement rate on wages. This system safeguards households against outliving savings, catastrophic medical expenditures and

many other risk items that can drive savings up in other settings, which substantially limits several standard precautionary motives for holding private wealth[1].

Moreover, the Dutch policy rules and administrative data provide the variation needed to quantify the two motives. For precautionary savings, the LTC copayment schedule features a statutory cap that generates a kink in care costs. This kink creates quasi-experimental variation in expected care costs across households that are otherwise similar. For bequests, the administrative data allow me to link households to their descendants and observe changes in the set of potential heirs. I study two such events: the death of an adult child and the birth of an unexpected grandchild. Conceptually, these events shift the marginal value of bequests in opposite directions, implying wealth responses of opposite sign under a bequest motive.

The data in this paper is provided by Statistics Netherlands (CBS) and covers the universe of Dutch residents. Crucially, there is detailed information on household wealth and family composition, allowing for linking all family members to one another. I merge various administrative registers at the household level, including civil register data, personal income tax data, information on wealth items, inheritance records and monthly sources of income. The analysis data is a yearly panel spanning almost two decades and contains detailed information on wealth and income of old age households, as well as life events of their heirs.

I begin by documenting the wealth patterns of Dutch retirees. Prior work has studied old-age wealth extensively, especially in the context of the U.S. Nevertheless, wealth trends in the Netherlands might be different due to different public insurance systems, and the results motivate the subsequent analysis. I find that wealth remains largely flat in old age: the average Dutch household draws down a mere 16% of their retirement-entry wealth over two decades of retirement. Theoretical benchmarks with conservative assumptions predict over 70% decline over the same period. While the findings are in line with prior work on the U.S., they are even more puzzling in the Netherlands, where the strong welfare state limits a variety of old-age expenditures.

In additional analysis, I look at different measures of wealth and compute household level distributions of wealth drawdown. Interestingly, excluding home equity does not change the findings. Both financial wealth, and wealth-net-of-home-equity produce the similar flat pattern, which suggests that housing cannot be a sole driver of the absence of dissaving. Furthermore, the absence of dissaving is very common in the Dutch population. The median household draws down only 10% of their wealth until death and around 45% of the population dies with more real wealth than they entered retirement with.

Next, I investigate how expected care costs affect wealth accumulation in a regression kink (RK) design, similar to Landais (2015), a variant of the regression discontinuity design that exploits a change in slope rather than a jump in levels. The design exploits the statutory cap in the LTC copayment schedule. The copayment depends on a means-tested measure of the household's income and wealth, which I call its contribution base. Below the cap, an additional euro of contribution base increases annual care costs one-for-one. Above it, the costs remain constant, which

[1] One exception is the long-term-care expenditures, which could still substantially alter the household finances.

creates a discontinuity in the slope of the schedule. The RK estimator compares the change in the slope of wealth accumulation at the threshold to the change in the slope of care costs (Card et al., 2015). This ratio identifies the marginal effect of care costs on wealth for households around the statutory cap.

I find sizable marginal responses for both singles and couples. An increase in the future care cost by €1 increases annual change in wealth by 45 cents for singles and 46 cents for couples. The bulk of the effect is driven by households in later stages of retirement. Compared to younger households, estimates are three to six times larger for households above age 80, which is in line with the higher probability to use institutional care in this age range.

Accumulating the estimated yearly response until the expected age of care entry, I calculate that singles and couples hold 22% and 21% of their wealth, respectively, in anticipation of future care costs. This calculation relies on two key assumptions: that every household expects to pay the copayment of a household at the cap, and that the marginal response applies to this amount in a linear fashion. In practice, households below the cap face lower care costs than the amount assumed in the calculation. I therefore interpret the estimate as an upper bound on the average precautionary share of wealth.

The final part of the analysis examines the desire to leave bequests. I exploit shocks to the set of potential beneficiaries for identification. I focus on two events: the death of an adult child and the birth of an unexpected grandchild, defined as a birth when both parents are over age 40. For each event, I use a matched difference-in-differences (DiD) design, pairing affected households with otherwise similar households that do not experience the event. I match households three years before the event based on a rich set of observable information including socio-economic status and pre-existing family structure. I then compare the evolution of wealth across treated and matched households before and after the event.

I estimate that households hold steadily less wealth following the death of an adult child, and increase their wealth following the birth of an unexpected grandchild. Losing a child leads to €11,600 less wealth five years after the event, or 5.5 percent of the mean wealth in this sample. The magnitudes are comparably smaller for the grandchild births: grandparents hold €3,400 more wealth than matched controls, which amounts to an increase of 1.2 percent. The opposite-signed responses are consistent with an intentional bequest motive directed toward likely beneficiaries.

To translate these responses into the total effect of bequests on wealth holdings, I scale the estimates by the number of children and grandchildren in the household. Taken at face value, this calculation attributes 17.5 percent of household wealth to an intended bequest motive, of which 14 percentage points are accounted for by children and 3.5 by grandchildren. This extrapolation is consistent with the heterogeneity results: households with two and three-or-more children show similar wealth responses to the loss of a child.

The results show that precautionary and bequest motives are quantitatively important but incomplete explanations of the retirement savings puzzle. Together, care costs and bequests can explain up to 40 percent of the observed mean wealth, narrowing the gap between the empirical

drawdown of 16% and theoretical prediction of above 70%. According to my estimates, the two prevailing motives are comparable in magnitude. Additionally, because each motive is estimated from a separate quasi-experiment, they operate at least partially independently in the sense that one is not a simple by-product of the other. For instance, if bequests were merely what is left over from saving against the cost of care, wealth would not respond to changes in the set of beneficiaries. These results highlight the importance of considering both mechanisms jointly in economic policy and models.

Moreover, the findings provide evidence in favor of an *intended* bequest motive directed towards a defined set of beneficiaries. Whether bequests are intended, accidental due to unexpected deaths, or incidental resulting from a by-product of precautionary savings has been a subject of debate in the literature (French et al., 2023). The events I study move the set of beneficiaries in opposite directions, and household wealth responds accordingly. This pattern is difficult to reconcile with purely accidental or incidental bequests, since wealth left over by chance or held against future care costs should not respond to who remain as potential beneficiaries.

The findings also speak to the form of the bequest motive. A preference defined over the size of the estate alone would be indifferent to the identities of recipients, whereas the findings track changes in the beneficiary set. At least part of the wealth held in old age therefore appears to be reserved for particular individuals. This result has direct consequences for the taxation of intergenerational transfers. For instance, for inheritance tax policy, this implies recipient-specific tax rates can have different welfare consequences compared to a tax on the size of the total estate.

This paper contributes to an extensive literature on saving behavior in old age (see De Nardi et al. (2016), Suari-Andreu et al. (2019), and French et al. (2023) for overviews). That retirees draw down their wealth slowly has been documented at least since Mirer (1979) and Hurd (1987). The two leading explanations are precautionary saving against medical and long-term-care expenses, and the desire to leave bequests. Life-cycle models with uncertain life expectancy that include both motives fit the retirement wealth profile well. For instance, De Nardi et al. (2010) estimates such a model by matching median wealth by age, cohort, and permanent income. Wealth data alone, however, cannot tell the two motives apart, as these models fit wealth almost equally well with or without a bequest motive (French et al., 2023). More recent work complements this approach with additional information from insurance choices, family structure, inter vivos transfers, cross-country variation, and inheritance decisions at spousal death (Ameriks et al., 2011; Lockwood, 2018; Banks et al., 2019; Ameriks et al., 2020; Kvaerner, 2023; Nakajima and Telyukova, 2024; Fella et al., 2024; De Nardi et al., 2025). The effect of each motive depends on which information is used and the quantification of each motive remains unsettled.

I make three main contributions to this literature. First, I provide novel quasi-experimental evidence on the two leading explanations of the retirement savings puzzle. Whereas previous work has largely relied on survey data and structural methods, I exploit policy variation in long-term-care costs and shocks to the set of potential heirs in reduced form designs. This allows me to estimate the marginal effect of each motive without relying on a structural model to disentangle

the two.

Second, I use the reduced form estimates to quantify how much of old-age wealth the two motives can account for at most. The extrapolations find that precautionary and bequest motives are comparable in magnitude: up to 21% and 17.5%, respectively. Together, the two key motives can explain a substantial portion of wealth late in life. Nonetheless, a significant share remains unexplained, pointing to additional factors that sustain wealth holdings in retirement.

Third, I extend the evidence that retirees decumulate little in the context of a generous European welfare state (Alessie et al., 1999; Van Ooijen et al., 2015; Suari-Andreu et al., 2019). In comparison to prior work, I document the retirement savings puzzle using population-wide administrative data over a long horizon. Administrative registers provide precise measures of household wealth and allow me to track the full population of households over almost two decades of retirement. At the same time, the Dutch welfare state insures households against many risks that could otherwise generate precautionary saving. I show that the retirement savings puzzle persists even in this setting.

This paper also connects to a growing literature on bequest motives, estate planning, and inter vivos transfers (Kopczuk, 2007; Kvaerner, 2023; Sturrock et al., 2025). These studies identify bequest-related behavior primarily from variation in mortality risk and transfers around the end of life. I instead exploit changes in the set of potential beneficiaries and investigate whether household wealth responds to who stands to inherit.

The paper proceeds as follows. Section 2 gives a brief overview of the Dutch institutional setting and describes the analysis data. Section 3 presents descriptive evidence on old-age wealth and establishes key empirical facts on old-age savings behavior. Section 4 examines the effect of precautionary savings on wealth accumulation behavior of Dutch households and Section 5 investigates the role of bequest motives. Section 6 concludes.

2 Empirical Setting and Data

2.1 Empirical Setting: the Netherlands

In order to provide context for the subsequent analysis, I briefly highlight some of the characteristics of the Dutch welfare state[2].

The Netherlands is one of the most generous welfare states in the world. It combines generous public insurance schemes that cover a variety of risks which old-age households face. Health insurance is universally covered through private suppliers, but is mandatory and suppliers are not allowed to vary prices. There is a deductible which amounts to €385 per year and a copayment on drug prescriptions for the first €250 of use. In total, households can expect to pay a maximum of €635 per year in the majority of the estimation window, when they make full use of the health system. Similarly, there is a universal public long term care insurance (LTCI), which covers home

[2]I discuss specific features of LTC and bequests in the later sections.

and institutional care. Copayments into the system are means-tested and importantly are capped at a maximum annual amount[3].

Beyond these health-related coverage, the welfare state features a generous pension system with the mean replacement rate of 75% (OECD, 2025b). The system primarily consists of a basic state pension, and mandatory occupational pensions. The state pension depends only on the number of residence years in the Netherlands. It is a flat rate which aims to provide a consumption floor in old age. Furthermore, employees are automatically covered by unemployment and disability insurance which award earnings-related benefits, protecting all laborers against these adverse shocks. Finally, the taxation of intergenerational transfers does not reward holding on to wealth until death. Gifts and inheritances are taxed under the same law and at the same progressive rates, but gifts to children benefit from an annual tax-free allowance, whereas the inheritance exemption applies only once. If anything, this encourages households to transfer part of their wealth during life, and there is recent evidence that Dutch households make use of these incentives before death (Sturrock et al., 2025).

This comprehensive welfare system coincidentally creates a suitable lab to study the savings puzzle. First, many risk items that drive savings upwards in the context of other countries are muted here. This makes high wealth at old-age even more puzzling, and allows me to isolate other mechanisms. For instance, the universal health insurance with near-zero copayment eliminates the medical expenditure risks that influence wealth accumulation in countries like the U.S., where these costs can in general be uncapped. Another example is the generous pension system: the mandatory mean replacement rate of 75% stands above many other countries[4] and insures households against outliving their savings.

Second, the public insurance schemes provide quasi-experimental variation that allows for testing some of the standard explanations. In particular, I exploit kinks generated by the public LTCI rules to test the effects of precautionary saving against long term care expenditures on savings motives. The universality of these schemes allow for methods that require precision and large samples. Third, the Netherlands has high quality administrative data on the universe of Dutch residents. Beyond enabling the analysis of the variation from insurance schemes, this further allows linking family members, detailed wealth and income information and inheritance registers together in large panel data sets. I make use of this rich data structure in analysing how life events to children can inform us on the parents' bequest motives in section 5.

Finally, while the Dutch setting might not be comparable directly with the U.S., it remains representative of many other developed economies. For instance, the universal public health insurance is common across OECD members, but in general not available in the context of the U.S.[5]

[3] In addition, there is very limited scope for quality improvements in health coverage through additional payments. For instance, patients have an unrestricted choice of providers irrespective of insurance status (OECD, 2025). Moreover, private insurance has historically provided little additional access to enhanced hospital amenities such as private rooms (Tapay and Colombo, 2004).

[4] see, for example, Figure 4.2 of the OECD (2025b)

[5] 26 out of 38 OECD countries achieve 100% health insurance coverage through public schemes or mandatory private suppliers, and 34 out of 38 achieve over 95% coverage (see, for example, figure 5.1 of the OECD (2025a)).

2.2 Data

I use administrative data on the universe of Dutch residents provided by Statistics Netherlands (CBS). I construct the main dataset by linking together a number of administrative registers at the household level. This includes civil register information, personal income tax data, information on wealth items, inheritance records, and monthly sources of income. The registers are linked through pseudonymised personal identifiers, which allows me to follow both retirees and their adult children over the full sample period.

The main outcome of interest in this paper is household net wealth, which is defined as the net private wealth of the household, excluding the pension wealth stored in the form of annuities. Wealth is reported at the end of each calendar year at the household level in the data, and is available from 2006 onwards. Therefore, I focus on the period between 2006 and 2024 unless otherwise stated. Finally, I adjust wealth for inflation if the analysis itself does not account for it. In particular, descriptive analysis in section 3 uses inflation-adjusted wealth using the 2015 base year.

The household is defined as the individual, and when available, their spouse. I require both the individual and their spouse to be born between 1920 and 1950. Furthermore, I require both the individual and the spouse to be at least 65 years old at panel entry. That is, an individual who is born in 1950 only enters the panel when they turn 65 years old in 2015. This process yields 39.3 million household-years and around 3.1 million households in the main estimation sample.

Panel A of Table 1 shows summary statistics of the main sample. The average household is 76 years old, and 71% of households have at least one child, with 1.9 children on average. Mean net wealth is €262k, or €137k excluding home equity, and mean financial wealth is €90k. Annual gross pension income averages €23.8k. Panels B and C split the sample into singles and couples, where couples include the years after the death of a spouse. Singles account for slightly more than half of all household-years and are less likely to have children (51% versus 94%). Couples hold more wealth (€320k versus €192k on average) and receive more pension income (€33.4k versus €15.6k). Appendix Table A1 reports the same statistics in the first year each household is observed: households enter the sample at age 70 on average, with mean net wealth of €270k. The estimation samples for the analyses in Sections 4 and 5 are described in the corresponding sections.

3 Wealth Profiles in Old Age

3.1 Drawdown of Wealth and Comparisons to Life-cycle Benchmark

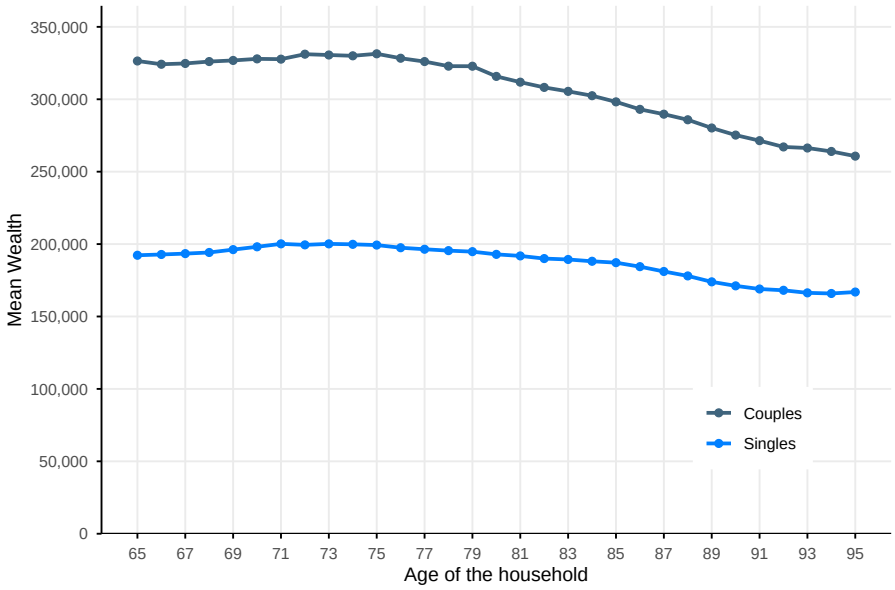
I begin by presenting wealth profiles in old age for Dutch households. Figure 1 presents aggregate mean wealth for Dutch households in retirement ages, separately for single and couples. Both single and couple household averages are largely flat across the old ages. This is especially seen until around age 80. In later years, the average wealth decreases, but this decrease is modest: for couples, where this trend is stronger, the decline amounts to less than a quarter of retirement-entry wealth.

TABLE 1: SUMMARY STATISTICS

	Mean	Std. Dev.	P10	P25	P50	P75	P90
Panel A. All households							
Average household age	76.2	6.9	67.5	71.0	75.5	81.0	86.0
Share with any child	0.71						
Number of children	1.87	1.68	0.00	0.00	2.00	3.00	4.00
Net wealth	262,034	1,690,968	1,623	11,799	96,979	310,692	576,934
Net wealth excl. home equity	136,524	1,656,234	1,121	7,110	26,287	84,132	257,166
Financial wealth	90,060	818,400	1,494	7,327	25,578	72,117	192,993
Pension income	23,786	19,953	1,112	14,250	20,692	31,003	45,108
Number of households	3,111,112						
Number of observations	39,313,948						
Panel B. Singles							
Average household age	76.5	7.3	67.0	70.5	76.0	82.0	86.5
Share with any child	0.51						
Number of children	1.35	1.71	0.00	0.00	1.00	2.00	3.32
Net wealth	192,117	873,500	817	5,563	31,019	234,630	479,214
Net wealth excl. home equity	103,100	825,691	612	4,080	19,317	64,820	212,367
Financial wealth	76,499	338,250	898	4,334	18,942	57,556	171,298
Pension income	15,618	14,402	0	2,823	14,910	21,508	31,617
Number of households	1,754,796						
Number of observations	21,012,800						
Panel C. Couples							
Average household age	76.0	6.5	68.0	71.0	75.0	80.5	85.5
Share with any child	0.94						
Number of children	2.47	1.45	1.00	2.00	2.00	3.00	4.00
Net wealth	319,551	2,115,801	4,297	23,623	172,238	361,711	649,342
Net wealth excl. home equity	163,603	2,082,833	2,201	11,036	33,276	100,347	296,649
Financial wealth	101,718	1,061,391	2,765	11,062	31,495	83,495	211,039
Pension income	33,400	21,321	16,886	21,315	28,327	39,728	55,342
Number of households	1,304,602						
Number of observations	17,843,078						

Notes. Summary statistics for the main sample: Dutch households in which all members are aged 65 or older, 2006–2024, one observation per household-year. Monetary values are in real terms, in 2015 euros (deflated with the consumer price index). Net wealth is total private wealth, excluding annuities; financial wealth is the value of financial assets. Pension income is gross annual pension income, summed over household members. Couples comprise married or cohabiting households and widowed households, i.e. including the household-years after the death of a spouse; the full sample in Panel A also includes divorced households. For confidentiality, each percentile is reported as the mean of the observations within one percentile of the quantile.

FIGURE 1: MEAN WEALTH BY AGE



Notes. Mean household net wealth, adjusted for inflation, by age, estimated from a household fixed-effects specification.

To quantify this wealth trend, I calculate the average drawdown of assets in old age using a simple fixed effects model:

$$\ln W_{it} \;=\; \alpha_i \;+\; f(s_{it}) \;+\; \varepsilon_{it}, \tag{1}$$

where W_{it} is inflation-adjusted net wealth of household i in year t,[6] s_{it} is years since retirement (household age less 67), α_i is a household fixed effect, and $f(\cdot)$ is a polynomial of s_{it}. The household fixed effects model allows for the identification of the profile from within-household changes over the retirement horizon, netting out level differences across households.

The fitted profile $\hat{f}(s)$ traces mean log wealth by retirement duration. The predicted share of retirement-entry wealth still held after s years is $\exp\big(\hat{f}(s) - \hat{f}(0)\big)$, so the implied cumulative drawdown is

$$D(s) \;=\; 1 - \exp\big(\hat{f}(s) - \hat{f}(0)\big). \tag{2}$$

The first column of Table 2 reports the results of this estimation with a cubic specification of years since retirement polynomial $f(s_{it})$. The survival-weighted annual dissaving rate over the whole age distribution is only 1.6% of assets. When evaluated at the mean age at death[7],this number translates into 16% drawdown of wealth. In other words, over the roughly two decades between retirement at 67 and the average age at death of 86, households draw down barely a fifth

[6]Because the specification is in logs, I exclude household-year observations with non-positive net wealth ($W_{it} \leq 0$), for which the logarithm is undefined. These account for a small share of the estimation sample and consist mainly of households whose debt exceeds their assets.

[7]Mean age at death is the expected age at death of a household entering retirement. I calculate this by integrating the survival probabilities from the Dutch period life table for 2023 published by Statistics Netherlands (Statistics Netherlands (CBS), 2024), as distributed by the Human Life-Table Database (Human Life-Table Database, 2026). The expectation over the sample ages is 86, i.e. $s = 19$ years into retirement, taking age 67 as retirement onset.

TABLE 2: OBSERVED WEALTH DECUMULATION VERSUS LIFE-CYCLE BENCHMARKS

	Empirically observed	Life-cycle model (deterministic lifetime)	Life-cycle model (uncertain lifetime)
Panel A. Average annual wealth decumulation across retirement ages			
Annual dissaving rate (survival-weighted)	1.6%	11.4%	10.3%
Panel B. Wealth decumulation by mean age at death			
Wealth drawn down between ages 67 and 86	16%	100%	78%

Notes. All columns describe the decumulation of private wealth held at retirement entry (age 67), normalized to one. Panel A reports the average annual decline in log wealth over the retirement-age profile, weighting each age by its relative exposure: observed household-years in the data and survival probabilities in the life-cycle models. The empirical rate is calculated from the fitted cubic log-wealth profile over its in-sample support. Panel B reports the cumulative share of initial wealth drawn down between ages 67 and 86, the mean age at death under the baseline Dutch mortality distribution. Because Panel A averages changes across retirement ages, whereas Panel B compares wealth at two specific ages, the two statistics are not mechanically proportional.
Both life-cycle benchmarks assume CRRA preferences, a 3% real return, $\beta = 1/(1+r)$, no bequest motive, and a constant real pension income equal to approximately 6.7% of initial private wealth annually. The deterministic benchmark assumes known death at age 86 and fully depletes private wealth. The uncertain-lifetime benchmark incorporates longevity risk with $\gamma = 4$ and imposes a no-borrowing constraint. The mortality calibration uses Dutch survival probabilities from the Dutch period life table (Statistics Netherlands, via the Human Life-Table Database).

of their initial wealth.

To evaluate how the observed drawdown reconciles with theory, columns 2 and 3 of Table 2 report predictions from two life-cycle benchmarks. Both describe a household who enters retirement at age 67 with one unit of wealth and runs it down optimally under CRRA utility, a 3% real return, and no bequest motive. I assume households are risk-averse[8]. In the benchmark analysis, wealth and a pension income which matches the sample statistics are the only assets available for consumption[9]. The model is described and derived in full in Appendix A.

In the canonical life cycle model, households can perfectly predict their age at death and only care about consumption. Column 2 reports the prediction under this *deterministic lifetime* benchmark: households dissave 11% of their assets per year and deplete their wealth fully by death. Although perfect foresight is a strong assumption, the same prediction holds whenever a fair annuity market is available, since annuities let households convert wealth into lifelong income and insure against outliving their assets. In practice, however, few households annuitize[10]. Absent annuities, a risk-averse household must self-insure against longevity risk by holding a precautionary buffer. Column 3 of Table 2 reports this uncertain lifetime benchmark. The household still draws

[8] I take ($\gamma = 4$) for the calibration in this table. De Nardi et al. (2025) estimate 3.7 imposing common preferences across singles and couples, and 3.3 when re-estimating on singles alone.

[9] The availability of the pension income reflects the institutional case of the Netherlands, as all households are automatically covered by the flat-rate public pension which aims to provide a consumption buffer in old age. I take pension income net of taxes as 6.7% of the initial wealth, calculated from the summary statistics of the estimation sample for comparability to the empirical decumulation rate.

[10] See the annuitization puzzle literature (Benartzi et al. (2011), among others).

down 78% of its wealth by the mean age at death, at 10% per year on average, far outpacing the empirically observed rates.

Table A2 considers several extensions to the life-cycle model. First, I consider households that are extremely risk-averse. The discrepancy between the empirical observation and the theory does not reconcile even at very high risk-aversion: a household with risk aversion parameter $\gamma = 10$ decumulates 61% of their assets by death. Second, I consider the case with no pension income. Even with no pension income to insure longevity, households draw down a staggering 57% of their initial wealth. Third, I consider a long-lived household, at the 90th percentile of the mortality distribution. Due to the increased lifetime, this household spreads their resources across more years, implying a dissaving rate at 7.8% per year, but even this stays well above the empirical estimate.

In summary, Dutch households draw down assets very slowly compared to benchmark life-cycle model predictions. The observed 16% drawdown is roughly a fifth of the 78% that longevity risk could account for under conservative assumptions of risk-aversion, and the gap widens further against the canonical benchmark.

Other Measures of Wealth

A critical assumption nested in the life-cycle models is that assets are fully fungible: a euro held in one asset category is interchangeable with a euro held in another. In practice, this might not hold, for instance, if markets are illiquid, or if households enjoy ownership of certain assets beyond their financial value. Housing wealth is particularly exposed to both channels. There are often substantial transaction costs in the sale processes of real estate (Engen et al., 1999). Furthermore, individuals might enjoy living in houses they own compared to being a renter, as they can more easily modify their living spaces to fit their needs (Nakajima and Telyukova, 2020; French et al., 2023).

In order to understand if the unwillingness to downsize housing equity is the main driver of the flat profiles, Appendix Figure A1 replicates the Figure 1 separately for liquid financial wealth in Panel (a) and for net wealth excluding home equity in Panel (b). The former measure consists of shares, bonds, bank balances and savings accounts, whereas I construct the latter by subtracting the value of owner-occupied housing equity from the net worth of households. Both of these alternative measures follow a similar pattern to net wealth in Figure 1: assets of singles are quite flat into old ages and assets of couples see a modest decline, but remain close to the initial values. These results demonstrate that while housing might be a conceptually important mechanism, the savings puzzle is also present for various asset categories.

Cohort and Mortality Effects

Two widely-known issues in the estimation of wealth profiles are cohort and mortality effects. These forces bias the wealth in opposite directions. Earlier-born cohorts accumulate less wealth in their working lives in a growing economy, and enter retirement with less wealth. Since these

cohorts are overrepresented at older ages in an unbalanced panel, they bias the profile *downward*. On the other hand, wealthier households on average live longer. The households who survive into very old ages are more likely to be composed of these households. This mortality effect biases the profile *upward*.

These effects are visible in the raw wealth means figure A2 and medians figure A3. Panel (a) of Figure A2 plots heterogeneity by birthcohorts. At any given age, earlier born cohorts are more often than not poorer. At later ages, these cohorts are oversampled, which pushes the wealth down. Panel (b) shows the opposite pattern: households who die at later ages have more wealth at any given age.

Both of these effects operate through two distinct mechanisms. First, both the cohort and the mortality cause *level differences* between households. A household who is born late or dies later has on average a higher level of wealth compared to other households. Second, the two effects change the *composition of households* at different ages. Since the wealth data is only available from 2006 onwards, earlier born cohorts are in the estimation support only at younger retirement ages. Furthermore, as households die at different ages, the wealth profiles in older ages rely on households who still survive in the data.

I conduct multiple remedies to account for these effects. First, the profiles in Figures 1, A1 and equation 1 include household fixed effects that control for the differences in levels. A household's birth cohort, its permanent-wealth level, and its eventual age at death are all time-invariant, and they are absorbed by the household fixed effects. Therefore, the profile and the average decumulation are identified from within-household changes in wealth as a household ages, not from the level differences between households who die young and who die old. This removes the part of the bias that operates through *levels*, the fact that long-lived households are simply wealthier at every age.

Even though the household fixed effects control for the biased levels, the composition of households still change across the age profile. In other words, the composition of households in the estimation support shifts across ages. The Figure A2 shows this effect by itself cannot explain the flat profiles. Panel (a) of this figure plots the raw mean wealth for different cohorts. Within each cohort, mean wealth follows a flat profile. Panel (b) plots mean wealth for different ages at death. Within each age at death group, households retain a substantial sum of their wealth. If anything, the households who die later decumulate faster. Overweighting these households should therefore bias the pooled profile towards more drawdown, understating the observed flatness.

Finally, I re-estimate the equation (1) on two subsamples with opposite selection. The *entry sample* keeps only households first observed at ages 67-70. Because the panel runs from 2006-2024, most of these households are not followed to death, so the profile is *right-censored*. It is dominated by the early-retirement years, when mean wealth is still flat or rising, and misses the end of life drawdown observed in the raw means Figure A2. The *death sample* is the mirror image. It keeps only households observed to die. Because the panel begins in 2006, most decedents had already retired by then, so it is *left-truncated*. The fitted profile is dominated by the steep terminal decline.

Crucially, because the fixed effects already net out the level channel, the two biases now operate only through which segment of the retirement horizon each sample observes, and thus push the estimated profile in opposite directions.

Table A3 displays the results for the two censored samples and the full sample. The full sample evaluated at the age 86 is the empirically observed wealth drawn down in Table 2. At this age, the two censored samples bound the rate of drawdown: entry sample fitted line predicts a drawdown rate of 10%, while death sample predicts 35%. In other words, tweaking the sample such that the composition affects the estimates in the worst opposite directions yield a lower bound of 10% and upper bound of 35% of drawdown rates. Even the worst-case upper bound is less than half the conservative life-cycle benchmark of 78%. Taken together, the results from this subsection show that the low drawdown of wealth in old age cannot be an artifact of the mortality bias.

3.2 Distribution of Wealth Drawdown

Having characterized the average wealth profiles, I now turn to the distribution of wealth drawdown across households. For all households that are observed both at an early retirement age and at death, I compute the share of retirement-entry wealth the households drew down by death:

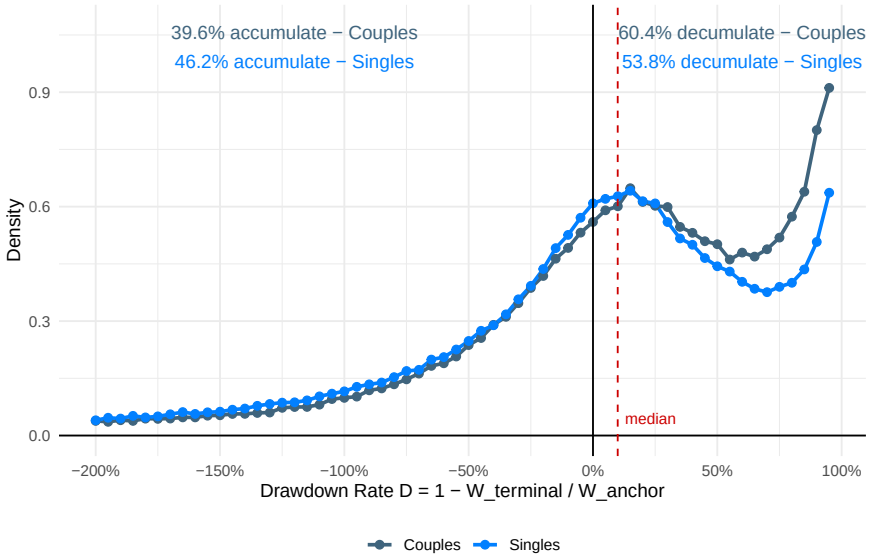
$$D_i = 1 - \frac{W_i^{\text{death}}}{W_i^{\text{initial}}}$$

where W_i^{initial} refers to the retirement-entry wealth at an age between 67–70, and W_i^{death} is the wealth these households had at the beginning of the year they died. I require that the time between death and anchor is at least three years to focus on households who has the time to drawdown resources.

Figure 2 plots the density of these realized decumulation rates D_i. Positive values denote decumulation and negative values denote the households who accumulated wealth over their retirement years. Three features stand out. First, both distributions are centered at low decumulation levels. The typical household draws down only modestly over the remainder of life, a median of 9.9%, so slow decumulation is broad-based in the sample. Second, there is a small share of households who decumulate most of the assets. This is more pronounced for couples. Since couples include widowed households, some of these might be attributable to side-bequests, i.e. the transfer of wealth at the first spousal death (De Nardi et al., 2025). Finally, a large share of households do not draw down at all. Among couples, 60% decumulate some wealth, while 40% leave at least as much real wealth as they began with. For singles, the split is closer to even, at 54% versus 46%. Accumulation of assets is very widespread in old age.

Because D_i requires observing a household both near retirement entry and near death, it is defined only on the sample of completed retirement spells. This sample over-represents households that die during the observation window relative to the full retiree population. To gauge how much this selection matters, Table 3 reweights the completed-spell median in two steps. First, Panel A applies inverse-probability-of-censoring weights (IPCW) that up-weights households whose spells

FIGURE 2: DISTRIBUTION OF REALIZED WEALTH DECUMULATION IN OLD AGE



Notes. Density of the realized decumulation rate $D_i = 1 - W_i^{\text{death}}/W_i^{\text{initial}}$, the share of retirement-entry wealth a household has drawn down by the end of life, by marital status. Wealth is measured in real terms. The vertical line at zero separates households that decumulate ($D_i > 0$, right) from those that accumulate ($D_i < 0$, left). D_i is defined only on completed spells, i.e. on households observed both near retirement entry and near death, and the spell is required to span at least three years between the entry and terminal observations so that early deaths do not mechanically register as zero change. The full-depletion mass at $D_i = 1$ and the small-denominator tails ($D_i < -2$ and $D_i \geq 1$) are trimmed for readability.

are more likely to be not observed at both endpoints. In practice, I run a logit model of being observed at both endpoints on predetermined covariates. The weights are the inverse of predicted probability of having a completed-spell (Robins and Finkelstein, 2000). A second correction, ω_a, additionally reweights the age-at-death composition of the completed sample to match the population age-at-death distribution[11].

Panel (a) of Table 3 reports the median of the household drawdown rates D_i. Before any weights are implemented, the median drawdown rate is 9.9%. Under IPCW and age-at-death weights ω_a, the composition-adjusted median drawdown rises to 15.4%, close to the 16% survival-weighted mean-profile drawdown from Section 3.1. In panel (b), I report the median drawdown rates by age-at-death groups. Median drawdown is modest for households dying early in retirement (1.9%), as well as in late seventies (8.8%); rises to 17.5% for those dying at ages 80-84, and reaches 29% for those surviving past 90. These rates match the average wealth profiles of Figure 1 and Figure 2(b), as both show flat patterns in early retirement, and modest decumulation in later ages.

The heterogeneity documented in this subsection sharpens the retirement savings puzzle. Standard life-cycle theory, even once longevity risk is taken seriously, predicts that virtually all households run down their wealth in retirement (De Nardi et al., 2016; French et al., 2023). A retire-

[11] Appendix A.4 details this weighting procedure.

TABLE 3: REALIZED WEALTH DECUMULATION ON COMPLETED SPELLS

	Median D_i	Households
Panel A. Selection correction		
Naive (completed spells)	9.9%	279,261
IPCW	7.2%	263,949
IPCW + ω_a	15.4%	263,949
Panel B. By age at death (naive)		
67–74	1.9%	79,187
75–79	8.8%	93,977
80–84	17.5%	77,673
85–89	20.5%	26,969
90+	29.3%	1,455

Notes. Median household-level realized decumulation rate $D_i = 1 - W_i^{\text{death}}/W_i^{\text{initial}}$ on the sample of completed spells, i.e. households observed both near retirement entry and near death with at least three years between the two observations. Panel A reports the median under alternative corrections for the fact that completed spells over-represent households observed through to death. *Naive* applies no correction. *IPCW* applies inverse-probability-of-censoring weights, which up-weight households whose spells are more likely to be censored before death so that the completed-spell median approximates what would be observed absent selective censoring (Robins and Finkelstein, 2000). *IPCW* + ω_a additionally reweights the age-at-death composition of the sample to the population age-at-death distribution. Panel B reports the naive median within bins of age at death. Dutch administrative population data.

ment in which 40% of couples and nearly half of single households instead preserve or build real wealth is difficult to reconcile with longevity precaution alone, and points towards motives under which retaining wealth is optimal. The remainder of the paper examines two of these motives that received widespread attention in the structural literature: precautionary savings against health-related expenditure risks, and bequests.

4 Precautionary Savings against Health-Related Expenditure Risk

The previous section documents little drawdown of wealth in old age, and that longevity risk alone cannot rationalize this behavior. Health-related expenditure risk can be a motive that fits this pattern well. Household healthcare consumption peaks at old age due to decreasing health. Literature from the US demonstrates that these expenditures can be catastrophic in old age. The effect of these expenditures is also easy to show in the life-cycle model: agents increase savings to make sure marginal utility of consumption is equalized across bad and good health states.

There are two main risk items: medical expenditure risk and care expenditure risk. As detailed in section 2, the Dutch medical system has almost zero expenditure items. As there is universal health insurance without price discrimination, this risk should not affect household decision making. There is also recent evidence that show wealth, if anything, increases following health shocks

and hospitalizations (Van Ooijen et al., 2015; Bonekamp and Wouterse, 2023). Even under worst case scenarios with maximum costs and long survival, the present value of maximum medical-related payments that households can incur throughout their old age is below 5% of mean wealth holdings, which further supports this assumption.

Apart from medical expenditures, long-term care (LTC) expenditure is the other primary risk factor for old age households. Care, especially in institutions, can be very costly. Many people also make use of it at some point in their lives. Due to diminishing health in old age, older households are particularly exposed to care needs. Therefore, the literature focused on the care expenditures as a potential source for precautionary savings in old age. The remainder of this section focuses on the effect of these care costs on savings behavior in a quasi-experimental design that varies care costs across households.

4.1 Setting and Identifying Variation

There is public long term care insurance in the Netherlands since 1967. The insurance covers home and institutional care, with eligibility criteria based on a care needs assessment. LTC usage requires a copayment, which is means-tested and is capped at a statutory maximum. Importantly, the public insurance is universal, and the private institutional care market is virtually non-existent in the Netherlands.

The central challenge in identifying the effect of care expenditure risk on savings behavior is finding quasi-random variation in the expected care costs. There are two main problems. First, in many settings, costs depend on endogenous choices such as prior private insurance takeup, and household characteristics that correlate with savings patterns, such as income and wealth. Second, household beliefs such as expectations and the likelihood to use care are unobserved to the researcher. Yet, these beliefs play an important role in the precautionary savings responses. For instance, a household that made an informal care arrangement with their children would have zero expected care costs attributable to public institutional care arrangements.

I get around these challenges by exploiting a kink in the Dutch public long term care copayment schedule. The copayment increases with the household's means-tested income and wealth up to a cap, and remains flat thereafter. Formally, the annual copayment a household faces upon institutional care entry is:

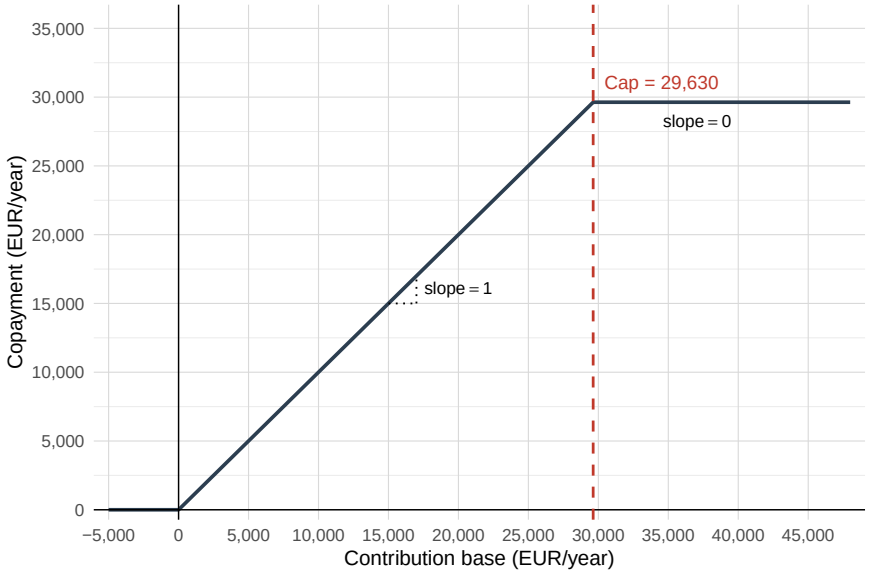
$$\textit{Copayment}_{it} \;=\; \min\Big\{\,\overline{C}\;,\;\; \max\big\{\,0,\;\; \underbrace{BI_{i,t-2} - 0.25\,(BI_{i,t-2} - f) - \textit{cvo} + \rho_w\,(w_{i,t-2} - \underline{w})}_{\equiv\; m_{it}}\;\big\}\Big\}, \quad (3)$$

where the base income $BI_{i,t-2}$ is net of the standard deductions,

$$BI_{i,t-2} \;=\; y_{i,t-2} - \tau(y_{i,t-2}) - z_{i,t-2} - k_i - a_i - 0.15\,l_{i,t-2}, \quad (4)$$

and y is household income, τ income taxes, z the health-insurance deduction, k a pocket-money allowance, a age-related deductions, and l labor income. In equation (3), f is a fixed exemption

FIGURE 3: LONG-TERM-CARE COPAYMENT SCHEDULE, 2021



Notes. Annual out-of-pocket long-term-care copayment (vertical axis) as a function of the household's means-tested contribution base m_{it} (horizontal axis), for the 2021 schedule. The copayment is zero below the exemption region, rises one-for-one with the contribution base, and is capped at the statutory maximum, generating a kink at the cap.

applied to base income, *cvo* a further statutory allowance, w is household taxable wealth, $\underline{w}$ the wealth exemption threshold, ρ_w the rate at which wealth above $\underline{w}$ enters m_{it}, and $\overline{C}$ the copayment cap. For any given year t, income and wealth from two years ago $t-2$ are taken into account. I refer to the inner argument m_{it} as the household's *contribution base*: income net of taxes, deductions and allowances, plus a small share of wealth above an exemption[12].

The rule creates three regions. Households whose contribution base m_{it} is non-positive pay nothing, an exemption that shields low-income, low-wealth households. Above the exemption the copayment rises one-for-one with m_{it}. Once the contribution base reaches the cap $\overline{C}$, the copayment is flat, so additional income or wealth no longer raises out-of-pocket care costs. The cap therefore creates a *kink*: the marginal copayment rate drops from one to zero where m_{it} crosses $\overline{C}$.

Figure 3 visualizes this copayment schedule for the year 2021. The x-axis denotes the contribution base m_{it}, and the y-axis the copayment that the household faces if it entered institutional care within the calendar year. The three regions are visible in the figure: below 0, and above $\overline{C}$, the slope of the copayment is flat, and between the two, the slope is 1. The dashed red line denotes the kink that I exploit.

The key idea is that below the kink, households to the right face higher future care costs and therefore should accumulate more precautionary savings. On the other hand, above the kink, households to the right face the same care cost, so any difference in the savings behavior cannot be attributable to expected care cost, but only in how the contribution base m_{it} itself links to the

[12] ρ_w is 0.08 of wealth above the exemption threshold for the copayment years 2017–2018 and 0.04 from 2019 onward. The cap $\overline{C}$ was €27,751 in 2017. It is updated yearly with respect to inflation and was €31,829 in 2023.

wealth accumulation patterns. In other words, a comparison of households below the kink would contain both the precautionary savings response to the expected care cost and how m_{it} is linked to the savings behavior, whereas above kink comparisons isolates the latter channel. The change in slope of savings behavior around the kink therefore identifies the precautionary savings response.

This setting provides two key advantages. First, the kink in the copayment schedule generates quasi-random variation in the *marginal copayment rate*: households just below and just above the cap face different slopes of copayment with respect to the contribution base but are otherwise comparable. Second, because the estimand is a reduced-form savings response to the schedule, households need not hold correct or homogeneous beliefs about their probability of needing care. The estimate captures how households adjust wealth in anticipation of the costs they would face if they use care, averaging over whatever beliefs they hold—precisely the object of interest for quantifying precautionary wealth.

4.2 Empirical Strategy: Regression Kink Design (RKD)

My primary empirical strategy is a regression kink design leveraging the kink created by the cap $\overline{C}$. I estimate the following regression kink model:

$$\Delta W_{it} \;=\; \alpha \;+\; \gamma_0 \left(m_{it} - \overline{C}\right) \;+\; \gamma_1 \mathbb{1}[m_{it} > \overline{C}] \left(m_{it} - \overline{C}\right) \;+\; \varepsilon_{it}, \qquad |m_{it} - \overline{C}| < h, \tag{5}$$

where $\Delta W_{it} \;=\; W_{i,t+1} - W_{i,t}$ is the change in household net wealth, the contribution base m_{it}, centered at the cap, serves as the running variable, $\overline{C}$ is the copayment cap at the schedule kink, and h is the estimation bandwidth. The coefficient γ_0 captures the slope of wealth accumulation in m_{it} below the kink. Finally, γ_1 is the RKD coefficient: the change in that slope at the kink.

The RKD coefficient γ_1 captures the causal effect of a marginal euro *decrease* in expected care costs on wealth accumulation. Because the marginal care-cost exposure is flat above $\overline{C}$, the kink isolates the savings response to the expected care costs, net of the direct association between the contribution base m_{it} and wealth accumulation that operates on both sides of the kink.

In order to identify this causal effect, I require the standard RKD assumptions. First, the running variable should be as good as randomly assigned locally around the kink. In particular, this requires that the running variable is not manipulated. This is a natural concern in this setting, as the running variable is computed from the household's wealth and income. There are two main reasons for why this is unlikely in this scenario. First, the gains from manipulation are small. Most retirement income consists of public and occupational pensions that households cannot adjust over the retirement years. On the other hand, lowering taxable wealth by €1 reduces the copayment by at most 4 cents (8 cents before 2019). Second, the household income and wealth that determine the running variable are measured two years prior to the current year. Therefore, a household that wanted to pay less if it entered care needs to forecast its contribution base two years into the future under parameters that are updated every year by law. Consistent with this, Tenand et al. (2026) find no evidence of strategic wealth reallocation after a 2013 reform that raised nursing home copayments for households with taxable wealth. Responding to the schedule, by contrast, does not

require knowledge of the rule itself. Households can learn the level of the copayment they would face from the official calculator of the CAK, which uses the figures from their tax return. I test for manipulation directly in Section 4.4.

The second RKD assumption is that wealth would evolve smoothly absent the policy kink. This assumption would be violated if another determinant of wealth accumulation also exhibited a slope change at the cap $\overline{C}$. Two features of the setting make this unlikely. First, to my knowledge, the cap does not coincide with any other tax bracket, benefit cutoff, or regulatory threshold in the Dutch system. Second, because m_{it} depends on each household's own income *and* wealth, the kink falls at a different point in the income distribution for each household. A confounder would therefore need to generate a slope change at every household's individual kink location, a coincidence that is difficult to construct.

I estimate equation (5) separately for single and couple households over the copayment years 2017–2023, the period for which I can determine the parameters of the copayment schedule, and no large changes to the schedule occurs. Appendix Table B1 reports summary statistics for the households in the headline samples. I restrict both samples to households who receive no wage or self-employment income. The headline estimates use household-years within €5,000 (singles) and €4,250 (couples) of the cap, which yields 353,352 observations for the single sample and 616,131 for the couple sample. Compared with the main sample, households near the kink are of similar age, 77 for singles and 75 for couples, but they are wealthier. The average single household is at the 71st percentile, and the average couple at the 59th percentile, of the wealth distribution of households of the same type aged 65 and over. The estimation sample is thus upper-middle class on average, but not confined to the very wealthy. Between 2017 and 2023, a non-negligible 13% of singles and 25% of the couples were above the cap in the population.

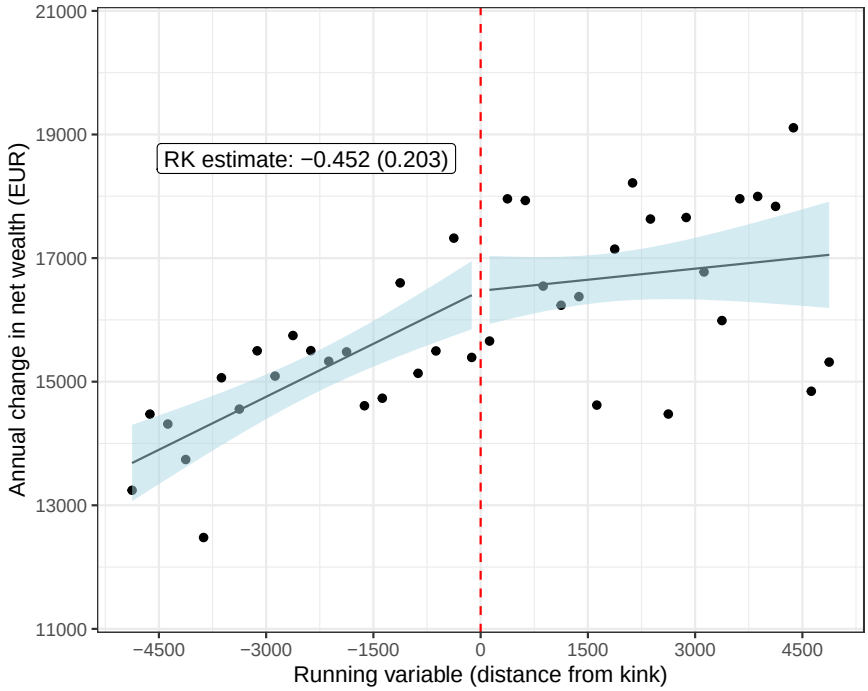
4.3 Main Results

Figure 4 presents the main results from the estimating equation (5), for both singles and couples. In both graphs, the running variable is normalized to be 0 at the copayment kink, such that $m_{it} = \overline{C}$, depicted by the red vertical line. The outcome of interest is the yearly change in wealth. The plots are restricted to the estimation bandwidths, which I determine using optimal bandwidth selection (Calonico et al., 2014) adjusted for the RKD estimation.

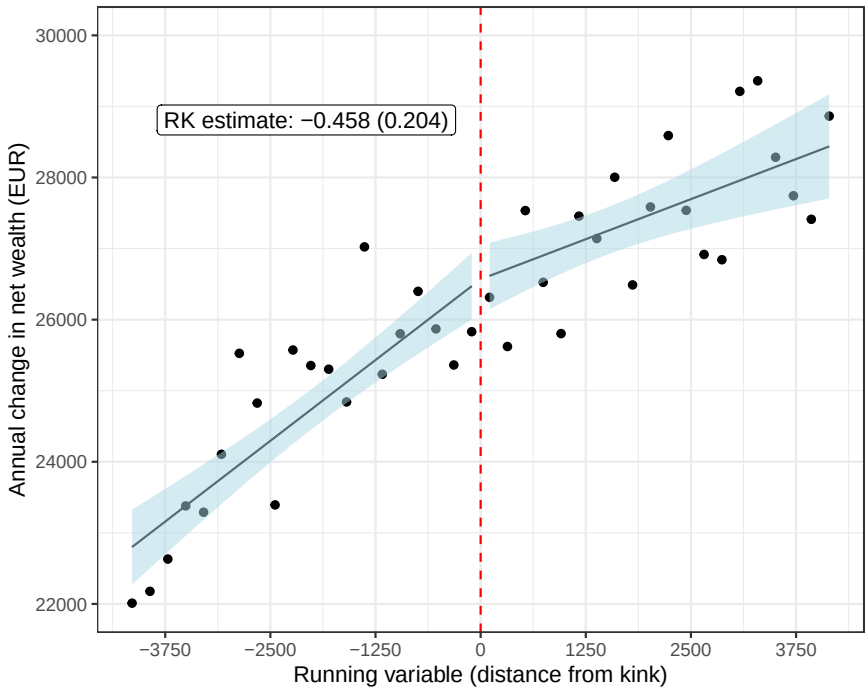
Panel (a) displays the results for the single households. Household wealth change is increasing in the running variable below the kink. The average single households saves around 13000 euros yearly at the lower estimation bandwidth, and 16000 euros around the kink. Above the kink, this relationship flattens out. RKD estimate reports the change in these slopes as -0.452, with standard error 0.203. This implies that singles save 45 cents for the marginal euro increase in future care costs around the copayment kink. This interpretation reads the negative of the RKD coefficient as the marginal savings response: since the marginal copayment rate falls by one at the kink, the slope change in wealth accumulation scales one-to-one into a response per euro of care costs. I defer the full discussion of this structural interpretation to Section 4.5.

FIGURE 4: SAVINGS RESPONSES TO CARE COSTS

(a) Singles



(b) Couples



Notes. Regression kink design estimates for single-person households (panel a) and couple households (panel b). The horizontal axis plots the running variable $m_{it} - \overline{C}$, the contribution base centered at the copayment cap. The vertical axis plots wealth accumulation ΔW_{it}. The solid lines are local linear fits on either side of the kink.

Couples results are presented in panel (b) of the plot. On average, couples save more of their wealth every year. The yearly change in wealth is 22500 at the lower bandwidth, and 27000 euros around the kink. The relationship between the running variable and the outcome is also more pronounced: the fitted line rises faster below, and stays rising above the kink. Nevertheless, the RKD estimate is very similar to the singles estimation. Couples retain on average 46 cents more on their wealth for the euro increase in future care costs.

To examine how care costs drive savings across different retirement ages, I run additional analyses for younger and older households. Appendix Figure B1 panels (a) and (b) report estimation results for singles below and above age 80, respectively. Due to the smaller sample size, results are noisier compared to the main estimation. Nevertheless, older singles respond significantly more strongly compared to younger households. RKD estimate is -0.717 for older singles and significant at 5%. On the other hand, RKD coefficient for younger singles is -0.261, and insignificant. This is despite the fact that wealth of younger singles increase faster: around the kink households increase yearly wealth by 20000 euros, double the wealth increase of older households. Panels (c) and (d) reports the analogous results for couples, which go in the same direction: the RKD coefficient is -0.232 for younger couples and -1.372 for older couples.

The heterogeneity indicates that age is an important element in the savings decisions against care needs. These results align well with the fact that the average age at entry into institutional care is 85 in the Netherlands (van Dijk et al., 2019). As households get closer to the ages at which they might need care, they respond more strongly in their savings behavior, consistent with households paying closer attention to care costs as the typical age of care entry approaches.

4.4 Identification checks and robustness

I present a wide range of robustness and specification checks for the RKD results. As discussed in section 4.1, estimating savings responses to care costs with an RK design requires two key assumptions: that households do not manipulate the running variable, and outcome would evolve smoothly absent the policy kink.

First, to test for the manipulation of the running variable, Figure B2 plots the corresponding densities of single and couple households used in the main estimation. Both densities are decreasing in the running variable, reflecting the skewed distribution of the income and wealth that make up the running variable. There is no observed bunching to either side of the kink, and no discontinuities visible in the densities that would pose a threat to identification of the RK estimates.

To elaborate based on the bunching framework, manipulation of the running variable would leave a distinct mark on these densities. Below the kink, each additional euro of the running variable raises expected care costs, so all households share an incentive to reduce it. Above the kink, marginal reductions save nothing: the only profitable deviation is a jump below the kink, where cost savings begin. Because adjusting income and wealth is costly, only households sufficiently close to the kink find such a jump worthwhile. In particular, there is a marginal deviator beyond whom no household moves. Manipulation would therefore create missing mass just above the

kink. The observed densities show no such gap: for both singles and couples, the density evolves smoothly through the kink, consistent with the absence of manipulation.

Second, whereas it is not possible to directly test the smoothness of outcomes absent the policy kink, it is possible to check the overall shape of the outcome using artificial kinks. The main concern to identification would be that change in yearly wealth would be a concave function of the running variable, and that the RK estimate picks up on this curvature. To check for the smoothness, I run the same estimation using two placebo kinks at $\overline{C} - 5000$ and $\overline{C} + 5000$. If the relationship between the outcome and running variable is concave, it is plausible that these placebos would pick up on the same functional form. Since there are no kinks in the copayment rules at these levels, they directly test this smoothness assumption.

Figure B3 and Figure B4 plots the results from these placebo estimates for singles and couples, respectively. None of the placebo results show a significant pattern. Visually, the placebos at the artificial threshold below the copayment kink looks linear. While the results at the above placebo threshold are noisier, they point to different directions for singles and couples, and also show no significant patterns. Taken together, these results suggest that it is very unlikely that main RK estimation results are affected by the relationship of the outcome and running variable.

I also test if a key predetermined variable, age, is smoothly distributed around the cutoff. Figure B5 shows the results from the RK estimation where I use age as the outcome. The slope change of age is 0. On both sides of the cutoff, age is flat at 77.3 for singles and 75.3 for couples, implying that households do not sort to be one or the other side of the kink based on their age. Since age is closely associated with health and care needs, this further suggests that manipulation is unlikely.

I report sensitivity checks to bandwidth selection in Figure B6. Panel (a) shows robustness to alternative bandwidth choices for singles. I mark the optimal bandwidth given by the CCT procedure (Calonico et al., 2014) with a "X". The main estimation for singles uses the closest bandwidth to this, 5000. I run the estimation using bandwidths from 4000 to 6000 and find that coefficients are quite comparable to one another. Panel (b) shows the analogous results for couples: while smaller bandwidths are noisier for this sample, the results are robust and point to the same direction once power is not an issue. Finally, I report the CCT optimal bandwidth selector results in B2. All estimates from these results point to the same direction.

4.5 Interpretation: from marginal flow response to precautionary stock

Section 4.3 showed that households accumulate around 45 cents per marginal euro increase in expected care costs around the kink. This statement rests on a structural reading of the RK estimate. By itself, the coefficient $\hat{\gamma}_1$ measures the change in the slope of the outcome, ΔW_{it}, at the kink. To convert it into the marginal savings response to care costs, I scale it by the change in the slope of the copayment schedule at the same point. As Figure 3 shows, the copayment rises one-for-one with the contribution base below the kink and is flat above it, so the policy slope falls from 1 to 0:

$$\hat{\beta}_{RKD} \;=\; \frac{\hat{\gamma}_1}{\Delta\!\left(\dfrac{\mathrm{d}\,\text{Copayment}_{it}}{\mathrm{d}\,m_{it}}\right)} \;=\; \frac{\hat{\gamma}_1}{0-1} \;=\; -\hat{\gamma}_1 \;\approx\; 0.45. \tag{6}$$

The denominator equals -1, so the structural parameter is simply the negative of the reduced-form coefficient: a €1 rise in the annual conditional-on-care copayment raises yearly wealth accumulation by €0.45 around the kink. Note that $\hat{\beta}_{RKD}$ is a reduced-form response: it absorbs household beliefs about the probability of ever entering care and the expected duration of a care spell. Households that consider care unlikely, or expect short spells, respond less to the same variation in the schedule, and this shading is built into the estimate.

The RKD estimate is a flow: extra euros saved per year, per euro of annual care cost. While this is informative as to whether care costs change households' savings behavior, it cannot inform on the precautionary stock of wealth that is held by the households. Section 3 documented that retired households hold onto their wealth rather than drawing it down, and the question is how much of that held wealth is precautionary saving against care costs. Answering it requires accumulating the yearly savings response over the years in which the household faces care risk and comparing the resulting stock to the wealth that households actually hold.

I conduct a simple back-of-the-envelope calculation in the following way: a household saves an extra β_{RKD} per euro of potential annual care cost in every year it survives until the expected age of care entry. Since the annual care costs are bounded by the copayment cap $\overline{C}$, applying this response to the copayment of a household at the cap gives the maximal annual flow. Accumulating this flow over the saving horizon yields the implied precautionary stock:

$$W_{\text{prec}} \;=\; \hat{\beta}_{RKD} \;\times\; \text{Care Cost} \;\times\; \sum_t \frac{S(t)}{(1+r)^t}, \tag{7}$$

Here, Care Cost is the maximum effective annual conditional-on-care copayment, $S(t)$ is the probability of surviving to year t of the saving horizon, and r is the real return. For ease of calculation, I assume the household is of the average age in the respective singles and couples samples, and I calculate the survival probabilities following this age.

This calculation relies on two key assumptions. The first one is linearity: the marginal response identified locally at the kink is extrapolated to every euro of the care cost. This allows for the multiplication of the RKD estimate with the Care Cost, taken as the effective maximum copayment at the cap. The second one is that household beliefs, such as care entry and duration, are on average similar for the estimation sample and the overall population. The RKD estimand β_{RKD} absorbs these beliefs for the estimation sample, the back-of-the-envelope carries them over to the average person.

Table 4 reports the results, and Appendix Table B3 the inputs. The calculation implies a precautionary stock of €71,200 for singles and €76,400 for couples, 22% and 21% of mean net wealth, respectively. Taken together, the calculations imply that precautionary saving against care costs accounts for roughly one-fifth of the net wealth that retired households hold. This is a substantial

TABLE 4: BACK-OF-THE-ENVELOPE PRECAUTIONARY WEALTH STOCK

	Singles	Couples
Precautionary wealth stock	€71,200	€76,400
Mean net wealth	€319,645	€359,590
Share of mean net wealth	22%	21%

Notes. Implied precautionary stock from equation (7): the RKD savings response times the effective annual copayment, accumulated from the mean age of each household type to the mean age at care entry (85), weighting each year by survival and discounting at a real return of 3%. Couples accumulate only while both spouses are alive. The share divides the stock by mean net wealth. Appendix Table B3 reports the inputs.

share of excess household wealth. Nevertheless, it leaves most of the wealth holdings documented in Section 3 unexplained, which motivates the analysis into different motives for savings, such as the bequest motive I turn to in Section 5.

Finally, the one-fifth figure is likely an upper bound on the precautionary share of wealth for two reasons. First, the calculation assumes that every household expects to pay the annual maximum copayment when in care, whereas households below the cap face lower conditional costs. Second, $\hat{\beta}_{RKD}$ is identified off households around the kink, who hold sufficient income and wealth to adjust their savings. The households with fewer resources might be less able to respond to future costs due to liquidity constraints or consumption floors. On the other hand, due to the cap imposed by the policy, the households around the kink are most exposed to the care costs compared to households with larger resources, who also face the same cost structure but can rely on a larger pool of assets.

5 Bequest Motive

Having quantified the precautionary savings channel against long term care risk, I now turn to the question of whether old age households retain wealth because they value transfers to beneficiaries at the end of life. I first outline the key identification problem and show in a simple model how shocks to the set of potential beneficiaries provide variation that addresses it. Then, I introduce the empirical strategy, a matched difference-in-differences design, and display the results. Finally, I discuss the implications.

5.1 Conceptual framework and identifying variation

The primary objective of this section is to quantify the causal effect of *intentional* bequest motives on excess wealth holdings in old age. The key challenge for identification of this effect is that bequest motives are not directly observed, as they originate from latent household preferences. Proxies based on realized bequests may reflect precautionary saving or other motives that drive savings behavior. Cross-sectional comparisons between households with different family structures can

also be misleading, as marital choices and family size are endogeneously determined by the same preferences that govern saving decisions.

I address these challenges by exploiting changes in the set of potential beneficiaries *within* households. I primarily focus on two life shocks: the death of an adult child and the birth of a grandchild. Focusing on the same household before and after these shocks keep the bequest preferences stable [13], but change the family structure over which this preference operates. If household values bequests, losing a potential beneficiary should reduce the marginal value of bequests, while gaining one should increase it.

To formalize this mechanism, consider a household that enters retirement with wealth W. The household allocates W between own consumption C, and total bequests B. Let N denote the effective number of intended beneficiaries. The household solves:

$$\max_{\{C,B\geq 0\}} \quad u(C) + \gamma V(B, N) \qquad \text{subject to} \qquad C + B = W, \tag{8}$$

where $\gamma \geq 0$ measures the strength of the household's bequest motive, and utility is increasing and concave in own consumption and bequests left to beneficiaries: $u' > 0, u'' < 0, V_B > 0, V_{BB} < 0$. The first-order condition (FOC) for the interior optimum is:

$$u'\left(W - B^*\right) = \gamma V_B\left(B^*, N\right) \tag{9}$$

I impose the following assumption:

Assumption 1. The marginal value of bequests increases in the effective number of beneficiaries:

$$V_{BN} > 0 \tag{10}$$

This assumption says that, holding total bequests fixed, households value an additional unit of bequests more when they have a higher number of intended beneficiaries. One intuitive reason is that having more intended heirs can increase the value generated by a given bequest. For instance, consider altruisticly motivated households who care about the well-being of their beneficaries. Because the benefit of additional wealth is diminishing for any one person, spreading a bequest across several beneficaries can generate more value than concentrating it on a single beneficiary. Similarly, a warm-glow motivated household can feel the joy from giving to more individuals.

Differentiating both sides of the FOC with respect to N and rearranging yields the key prediction:

$$\frac{\partial B^*}{\partial N} = \frac{\gamma V_{BN}\left(B^*, N\right)}{-u''\left(W - B^*\right) - \gamma V_{BB}\left(B^*, N\right)} > 0, \quad \text{and} \quad \frac{\partial C^*}{\partial N} < 0 \tag{11}$$

where the strict inequality follows from Assumption 1, concavity of u and of V in its first argument, and the assumption that the household values bequests, $\gamma > 0$. Since total wealth must be split between own consumption and bequest, gaining a beneficiary must increase the desired

[13] to the extent that these preferences are unaffected by shocks to children, discussed further in section 5.2

bequests, and reduce resources allocated to own consumption. Conversely, losing a beneficiary lowers desired bequests and increases own consumption.

To summarize:

$$\text{Losing a child} \implies N \downarrow \implies B^* \downarrow \implies C^* \uparrow \implies \text{lower old-age wealth}$$

$$\text{Gaining a grandchild} \implies N \uparrow \implies B^* \uparrow \implies C^* \downarrow \implies \text{higher old-age wealth}$$

The model as outlined above is intentionally agnostic about the precise bequest motive[14]. It is compatible with warm-glow, altruistic, dynastic and exchange-based motives insofar as changes in the set of beneficiaries affect the marginal value of resurces reserved for bequests. This generality is useful in the subsequent empirical analysis, as the primary object of interest is the total effect of bequests in this paper. The estimates should therefore be interpreted as evidence on the importance of intentional bequest considerations broadly, rather than as distinguishing among competing theories of bequest behavior.

As hinted above, many forms of bequest preferences naturally satisfy Assumption 1. There are two simple examples that can be interpreted as microfoundations: warm-glow directed towards heirs and altruistic bequests. Due to the concavity of the preferences, in both models having more heirs crowds out bequest transfers to existing beneficiaries, but increases aggregate bequests in the interior solutions. Intuitively, the warm-glow households enjoy making more bequest transfers in smaller sizes than making one big bequest, perhaps because they primarily enjoy the feeling of giving to a person. This would be similar to the *love of variety* a la Null (2011), which focuses on transfer motives to charities.

The altruistic foundation allows for a test of the mechanism behind the intended bequest motive. Permanent shocks which affect a child's income capacity should alter transfers during lifetime and at the end of life. To proxy for a negative permanent shock, I further look at the effect of persistent and sudden disability insurance take-up in additional results.

For the interpretation, two features of the framework stand out. First, N denotes the effective number of intended beneficiaries, not the number of legal heirs. A family member enters N to the extent that the household values leaving resources to that person, either directly or indirectly. A new grandchild may therefore increase N even without becoming a direct heir, provided that the birth raises the household's marginal value of bequests. Second, interpreting the resulting changes in wealth as evidence of a bequest motive requires that the events do not affect wealth primarily through other channels or substantially alter the underlying preference parameter γ. The empirical analysis therefore implicitly assumes that such effects are second-order, relative to the change in the marginal value of bequests, conditional on the design choices detailed in the next subsection.

The conceptual framework yields two predictions for observed wealth: households should draw down assets more rapidly after losing an intended beneficiary and more slowly after gaining one. The next subsections detail the empirical strategy and tests these predictions in the data.

[14] Apart from assuming that the bequest is intended, that is, not accidental, and directed towards a finite set of beneficiaries.

5.2 Empirical Strategy

5.2.1 Matched Sampling Procedure for Control Group Selection

A challenge in this setting is to find appropriate control groups across the different life events I study. I follow a growing list of other papers that employ a matched design in settings that employ life events as a source of identification (Azoulay et al., 2010; Rabaté et al., 2025; Jäger et al., 2026). For each treated household, I identify a placebo household whose child did not experience the event in the estimation time window, but who are otherwise similar in terms of lagged observable characteristics.

Let t denote calendar year, t_p^* denote the event (child death or grandchild birth) year of pair p. $k(i,t) = t - t_{p(i)}^*$ is the event time of household i, who is matched in pair p. Outcomes are measured at the beginning of the year, as wealth for year t is the end-of-year wealth from the previous period. Therefore, normalization is done with respect to $k = 0$, and $k = 1$ is the first period in the post-event period that outcome can respond.

The treatment group consists of households whose heirs had one of the three life events: child death, grandchild birth or child disability. In order to identify these households, I link parents to children and grandchildren, import birth and death dates for all heirs, and construct a yearly panel that includes information on marital history, age and sources of income for the children. For each event and event year, I sample control group from other households who never experienced that event.

The matching procedure is 1:1 exact matching without replacement, following matched-control event-study designs such as Azoulay et al. (2010) and Jäger et al. (2026). I operationalize it as follows: every treated unit gets an anchor period three years prior to the event. Then, I partition the data into cells based on matching variables and this anchor date. The control pool for a household is other households who are not treated under this event definition within that cell. For the matching, I use the following list of variables:

- Household characteristics at $t_p^* - 3$: birthyear, birthyear of the spouse (if not single), marital status, number of kids, number of grandkids, wealth and income deciles of the household within the anchor year, the age band of the eldest child (multiples of five)
- Calendar year of the event

For every treated household, I shuffle the entire control candidate list uniformly at random, and accept the first pair if and only if neither side has been used yet. If not, I move onto the second pair and iterate until all control candidates are exhausted. This procedure is attractive, as it searches for the maximal number of pairs while randomizing the control pool selection. Finally, every matched control inherits the treated unit's event year and both sides get trimmed to the estimation window.

This matched sampling procedure ensures that within a pair, the treated and control units share the same household age and same calendar years.

Table C1 reports the total number of treated households and number of households that could be matched to another pair. Out of the 49,247 treated households in the child death estimation arm, 48.5% could be matched to a suitable comparison group. The match rates for the grandchild birth and disability arm are 36% and 44% respectively. The low matching rates in the grandchild arm could be due to the restriction on the control group that the control does not experience a grandchild birth event themselves during the panel window. This might make it more difficult to obtain suitable comparison units compared to the rarer events of child death and disability insurance take up.

Sample Restrictions.

Beyond the global restrictions from Section 2, I require that the age of the child who experiences the shock (or becomes the parent to the grandchild) is between 25-57, and that households have less than five children ever born. The former restriction is there to focus on life events to children still in their working phase of their lives. The latter serves multiple purposes. First, the control pool does not have a child who experienced the event by construction. With many children, this restriction might lead to a sort of negative selection, as the likelihood that none of the children ever experiences the life event becomes smaller with more children. Second, many children families might have different sorts of family structures, such as drawing more heavily from previously separately married individuals who got married together in old age, but who do not necessarily have bequest motive towards their step-children. Since I count the total children of the household without differentiation based on this, I don't consider these families.

In cases where one household experiences multiple events, I take the earliest qualifying event. I also trim the households whose wealth changed in the top or bottom 0.1% in any year. This is to ensure primarily to exclude households who moved large estates into or out of the country, or who have measurement error in their wealth information. Furthermore, the main estimation requires that both the treatment and the control households are observed for at least three periods following the event, so that the effects are not driven by selective attrition. This causes a decrease in the estimation sample sizes substantially: for instance, table C1 shows 23,887 pairs were found in the child death arm, but only 13,193 survive this criteria.

Finally, I impose some important event-specific restrictions. For the child death event, I require that the deceased child did not have a child of their own. This is to ensure that the set of beneficiaries decreases in the child death event, as otherwise the burden of care of the surviving grandchild might fall onto the household. For the grandchild birth event, I require that the parents of the grandchild were both above 40 years old, to proxy for the grandchildren that were unanticipated. Finally, in the child disability onset, I require that the child was previously employed for two years and that they stay on the disability insurance for at least two years. This allows me to identify children who had stable career trajectories, but plausibly had a sudden disability shock that was absorbing.

Summary Statistics and Balance

Appendix Table C2 reports summary statistics and covariate balance at the matching anchor for the three event arms. Panel (a) reports the means for the matching variables. Because of the exact matching, these variables are identical across the treated and control units within each estimation arm. The average household in the child death estimation is 72 years old, has 2.5 children and 2.3 grandchildren. They are from the fifth wealth and income deciles. The households in the grandchild birth are slightly younger, at 68.1 years old, and richer, from the sixth income and wealth deciles. The disability arm households average 70 years old and are comparable to the child death arm in terms of wealth and income.

Panel (b) reports the means of covariates not matched on: mean age of children, net wealth and gross income. These variables look quite balanced. For instance, a worry in the child death event could be that within each wealth decile, households who experience a child death event are poorer. This is not the case: the treated households average €213,500 and controls €209,400, and are quite comparable on their income. The only small discrepancy comes from the mean age of children in the grandchild birth arm. Treated households have younger children by 0.7 years, however, this should not be a concern as the eldest child and household ages are identical by design.

5.2.2 Estimating Equation

I estimate the wealth response to a life event with the interaction-weighted estimator of Sun and Abraham (2021), and report the two-way fixed effects and the pair-by-event-time specifications in Appendix.

I index cohorts by the event year. A treated household i matched in pair p belongs to cohort $e_i = t_p^*$. Its matched control never experiences the event during the panel window, so I assign it $e_i = \infty$ and it enters as a never-treated unit. The first stage is the saturated regression

$$Y_{it} \;=\; \alpha_i \;+\; \lambda_t \;+\; \sum_{e} \sum_{k \neq 0} \delta_{e,k}\, \mathbb{1}[e_i = e]\, \mathbb{1}[k(i,t) = k] \;+\; \varepsilon_{it}, \tag{12}$$

where Y_{it} is the net wealth of household i measured at the beginning of year t, α_i is a household fixed effect and λ_t is a calendar year fixed effect. The double sum runs over event-year cohorts e and over event times k in the estimation window, omitting $k = 0$. Never-treated households contribute no interaction term; they identify λ_t and so supply the counterfactual calendar-year path. The coefficient $\delta_{e,k}$ is the average effect at event time k for the households whose event occurred in year e.

The reported event-study coefficients aggregate the $\delta_{e,k}$ across cohorts,

$$\beta_k \;=\; \sum_{e \in \mathcal{E}_k} \omega_{e,k}\, \delta_{e,k}, \qquad \omega_{e,k} \;=\; \widehat{\Pr}[\, e_i = e \mid e_i \in \mathcal{E}_k \,], \tag{13}$$

where $\mathcal{E}_k$ is the set of cohorts observed at event time k and $\omega_{e,k}$ is the share of cohort e among the treated households in that set. The weights sum to one and are non-negative by construction, so

β_k is an average of cohort-specific effects rather than a difference of them.

The estimation window is $k \in \{-3, \ldots, 5\}$. Coefficients are normalized at $k = 0$, which is the last period whose outcome is measured before the event can move wealth. The coefficients β_{-3} to β_{-1} are pre-event coefficients and β_1 to β_5 are the post-event effects.

The outcome Y_{it} is household net wealth for the grandchild birth and child disability takeup events, and net wealth less the cumulative inheritances the household received since panel entry for the child death event. This adjustment in the child death arm controls for the deceased child's estate flows to the household, which would otherwise bias the household wealth upwards following child's death. In Appendix plots, I also report the unadjusted wealth for this event.

I cluster standard errors at the pair level in all specifications. One-to-one matching without replacement creates dependence between the two members of a pair, and clustering at the level of the matched set remains valid when treatment effects are heterogeneous (Abadie and Spiess, 2022).

Identification Assumptions and Potential Threats. To identify the causal effect of life events to heirs, I require that these life events are exogeneous conditional on covariates included in the above procedure. In particular, this implies the parallel trends assumption, that household wealth would have evolved in parallel to each other in the treatment and control groups if the households had not experienced the life event. Since matching is done three years prior to the event, and not based on trends, the pre-event coefficients can be used to test for the validity of this assumption.

There are a couple potential threats. To start with, anticipation is a concern here for all event arms. Death of children can be anticipated in cases with long illnesses, grandchild births are potentially anticipated well in advance due to marital choices of children, and disability insurance takeup can be preceded by chronic conditions or long lasting illnesses. While I cannot rule out these channels completely, I conduct multiple remedies that should alleviate this concern. First, I require that the children are between 25-57, where deaths are relatively uncommon. Second, I restrict grandchild births to cases where both parents are above 40 years old. This is an age range that is relatively rare for parents to have children, and when grandparents probably have low expectations about new grandchildren. Third, I require that the disability insurance (DI) take-up is preceded by stable employment histories (at least ten working months in each of the preceding two years), and that the child uses DI as the primary source of income at least for two consecutive years. This helps isolate the disability cases that are more likely to have a sudden onset, and change the life quality of children permanently.

A further potential issue is the exclusion restriction. The effect I estimate possibly combines plausible confounds, such as the loss of an informal caregiver and funeral costs in the child death arm, and inter vivo transfers in the grandchild birth arm. I assume these effects are second order to the effect through the bequest motive. Furthermore, both confounds push wealth down in both cases, so cannot jointly generate the opposite-signed responses the framework predicts. The symmetry (or lack of) is a test of this and a reason to study the events together.

5.3 Results

Main Results

Figure 5 presents the results of the estimating equations 12 and 13 for the child death event and grandchild birth event. In both panels, the horizontal axis is the event time, the vertical axis is the estimated difference in wealth between treated and matched control households, and the reference year $k = 0$ is normalized to zero.

Panel (a) reports the estimates from the death of an adult child, estimated on the 13,193 matched pairs. From the year after the death onward, treated households hold steadily less wealth than their matched controls. The gap opens at roughly €6,900 in the first year and widens to €11,600 by the fifth (std. error 1,756), or 5.5 percent of the €210,000 mean wealth in this sample. While the first year drop is the most sizable, the path in the following years shows a monotone decrease in wealth, which aligns well with a drop in the desired bequests.

Panel (b) reports the results from the grandchild birth estimation over 17,179 matched pairs. The DiD estimate is positive: treated households accumulate more wealth than their controls after the birth. The gap is indistinguishable from zero in the first year, becomes significant from the second onward, and reaches €3,400 by the fifth (std. error 1,599), or 1.2 percent of the €289,500 mean wealth in this sample. The estimates are smaller and less precisely determined than in panel (a). As a grandchild is a more distant claimant than a child, the birth of a grandchild might not shift the beneficiary set as much as a child death event.

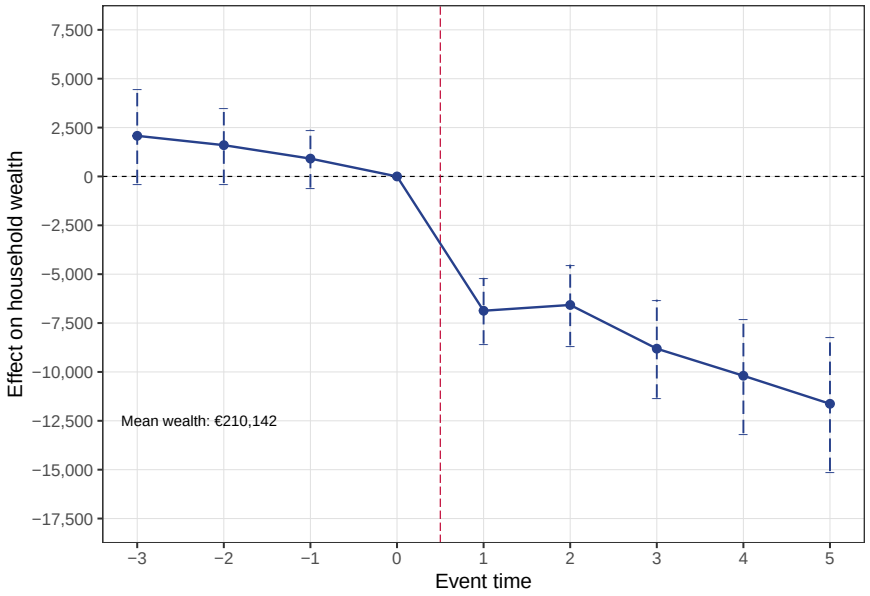
Both results move in the direction that the framework in Section 5.1 predicts. Under Assumption 1, losing a beneficiary lowers the marginal value of bequests and releases resources to own consumption, while gaining one does the reverse. The two panels deliver exactly this pattern of signs, which suggest the presence of an intended bequest motive directed towards children and grandchildren as one of the reasons for higher wealth holdings in old age.

Estimation Checks. I report the full estimation coefficients in Table 5. The coefficients in the pre-period in both plots are all individually insignificant. Nevertheless, there is a slight declining trend in the coefficients of the child death arm. In order to visually check pretrends, I plot time series of wealth for both events in Appendix Figure C1. Panel (a) reports the mean household wealth for the treatment and control group in the child death event, and panel (b) reports the analogous plot for grandchild births. There is a slight decrease in wealth prior to the death event, which would be in line with anticipation effects, especially from period -1 to 0. However, this drop is small compared to the subsequent widening of the gap following the child's death. This suggests that the anticipation might be second-order compared to the effect through the bequest motive.

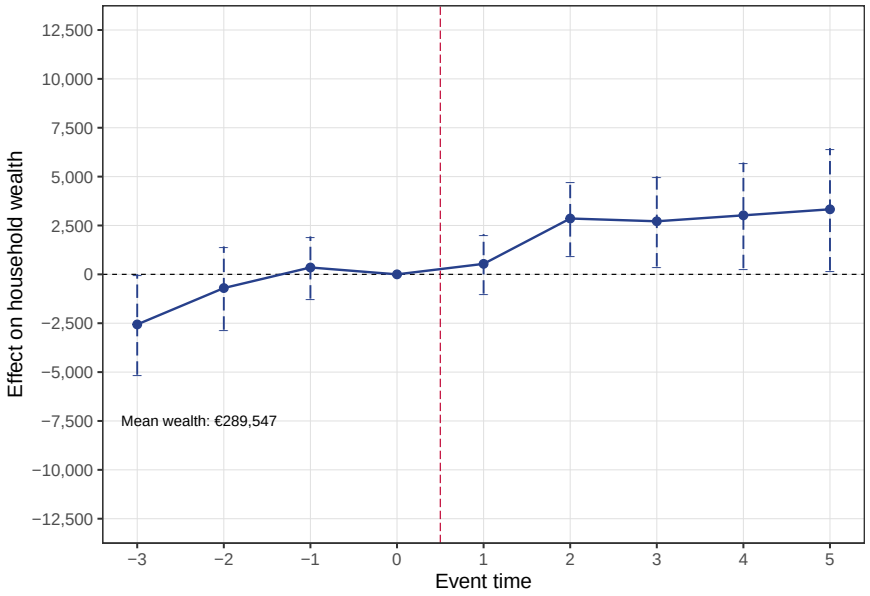
In the grandchild birth arm, event time -3 coefficient is only marginally insignificant. However, the trends in panel (b) of Appendix Figure C1 seem largely parallel, apart from this matching year, where the gap is slightly narrower. Control and treated households track each other well until the birth of the grandchild, and after the event, the control households stay on their declining trend,

FIGURE 5: WEALTH RESPONSES TO CHANGES IN THE BENEFICIARY SET

(a) Death of a child



(b) Birth of a grandchild



Notes. Matched difference-in-differences estimates of (12). The horizontal axis is event time, with the event year $k = 0$ normalized to zero. In panel (a) the outcome is household net wealth net of cumulative reverse bequests; in panel (b) it is household net wealth. Vertical bars are 95 percent confidence intervals from standard errors clustered on the matched pair.

TABLE 5: MATCHED DIFFERENCE-IN-DIFFERENCES ESTIMATES

	Death of a child Wealth, net-reverse inheritance	**Birth of a grandchild** Wealth	**Child enters disability** Wealth
	(1)	(2)	(3)
$k = -3$	$2,078^{*}$	$-2,562^{*}$	629
	$(1,236)$	$(1,313)$	(548)
$k = -2$	$1,601$	-701	-77
	(989)	$(1,088)$	(444)
$k = -1$	913	351	-193
	(755)	(812)	(326)
$k = 0$	0	0	0
$k = 1$	$-6,870^{***}$	537	-830^{**}
	(858)	(777)	(332)
$k = 2$	$-6,574^{***}$	$2,857^{***}$	$-2,205^{***}$
	$(1,053)$	(974)	(435)
$k = 3$	$-8,810^{***}$	$2,715^{**}$	$-3,434^{***}$
	$(1,280)$	$(1,180)$	(523)
$k = 4$	$-10,195^{***}$	$3,019^{**}$	$-4,893^{***}$
	$(1,500)$	$(1,379)$	(633)
$k = 5$	$-11,630^{***}$	$3,328^{**}$	$-6,809^{***}$
	$(1,756)$	$(1,597)$	(752)
Mean of outcome	$210,142$	$289,547$	$198,201$
Observations	$225,034$	$298,696$	$1,069,057$
Matched pairs	$13,193$	$17,179$	$62,173$
Pre-trend test (p)	0.883	0.171	0.678

Notes. Table reports coefficients from interaction-weighted-estimator 12 and 13 for the three event arms. The first column is results from the child death event, the second grandchild birth event and the third the child disability event. Standard errors are in parenthesis. The last four rows report the mean of the outcome for the treated units at the matching anchor year, number of observations, number of matched pairs and results from a Wald test on the joint significance of the pre-period coefficients.

whereas the treated household wealth stays flat, widening the gap between the pairs. Because the gap seems to be contained to the matching period, and not reflecting a general different pattern, it seems like this is an isolated coefficient rather than evidence of a differential trend between the treated and control units.

To formally assess the pre-trends, I test for and reject the joint significance of the cohort-specific pre-event coefficients across the three treatment arms. The p-values are reported at the end of Table 5.

Alternative to the interaction-weighted estimator, I run the standard difference-in-difference estimator with household and event-time fixed effects, and a second model with household and pair x event-time fixed effects that only make use of the variation between the matched pairs. Figure C3 displays the results from these models. The two alternative specifications track each

other very closely, as would be expected under the exact matching design with balanced panel. Both alternate specifications are very similar to the results from the main estimating equation in Figure 5.

Finally, in order to gauge if selective attrition drives the results, appendix table C3 reports the attrition rates for all events. I require that both the treated and control units are observed throughout the pre-event period as well as three years in the post period. Nevertheless, the post period coefficients of 4 and 5 can be biased due to selective exit of treated or control households. The results indicate that this is likely not an issue. Around 80% of households in the child death arm, 93% of households in the grandchild birth arm and 85% in the child disability arm are observed throughout the entire window. Across all events, both these shares, and the mean number of post-event years are essentially the same across the treated and control units.

Heterogeneity by Number of Children Next, I investigate the heterogeneity by number of kids in the impact of child death event. A natural question to ask in this setting is whether the loss of a marginal child has a bigger impact in smaller families. While these families lose a larger share of their potential beneficiary set, they might also have different preferences compared to larger families.

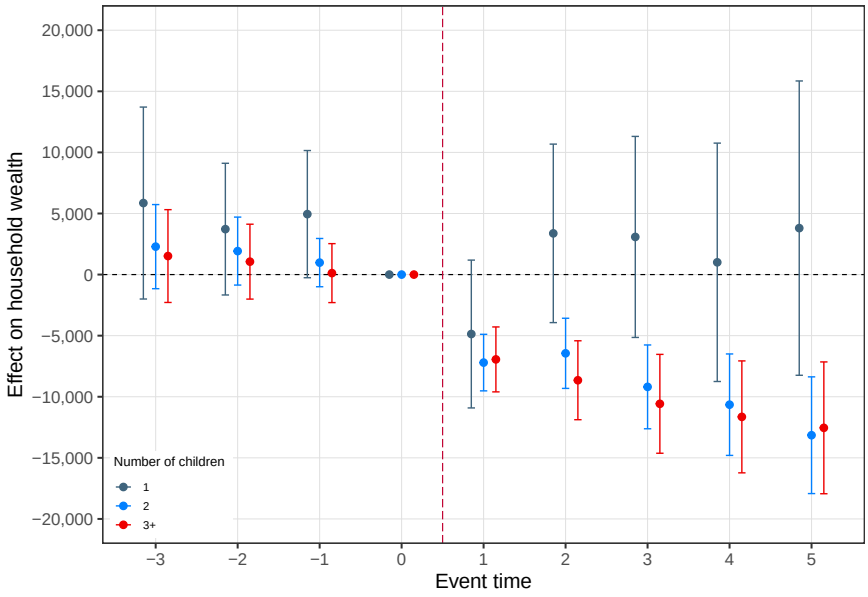
Panel (a) of Figure 6 displays the heterogeneity results by number of children, separately for one, two and more than three child families. The many child families drive the result. Results on the two and more-than-three child families track each other closely, which goes against the explanation that households with lower number of heirs are affected more. This might be due to the different preferences that these households have. Results on the one child families are uninformative. While there is a comparable drop in terms of the estimate in the first event year, the standard errors are too large to make inference in general. As Appendix Table C1 shows, these coefficients are estimated-off only 1,141 households, less than one fifth of the other two groups. Finally, panel (b) of the Figure displays the analogous results for grandchild births. Similar to the child death events, estimates on the one-child families are too underpowered for inference. The two child families show the strongest effects, and there is a smaller suggestive effect on the three child families. Table C4 summarizes these findings.

Disability event as a test of Altruism. As discussed in section 5.1, the estimation can speak to the effect of intended bequests broadly, and is silent on the precise bequest motive that operates underneath. Nonetheless, it would be informative to examine which mechanisms could be underlying the effects estimated in the main results.

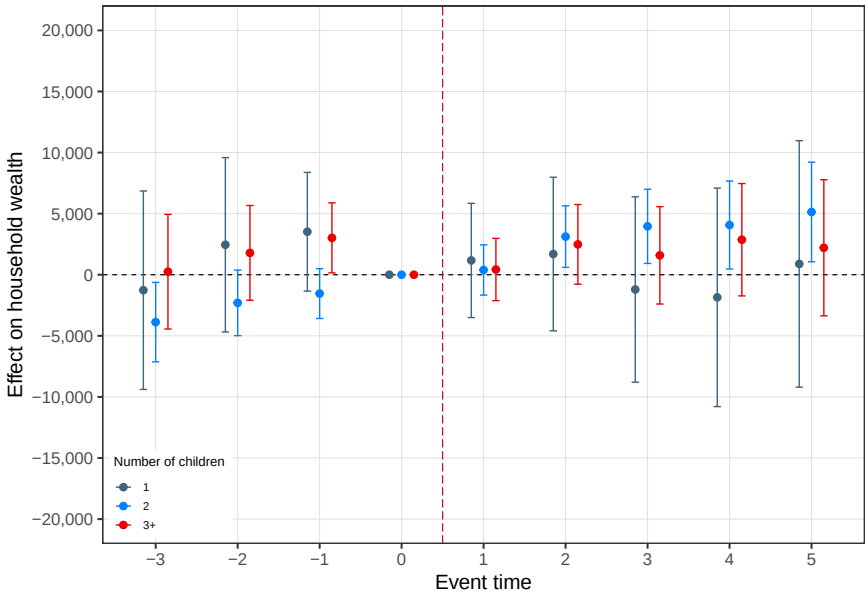
I attempt to uncover whether household wealth is affected by altruistic motives in old age by evaluating a third event which shifts lifetime earnings capacity of a beneficiary: disability events. Using the existing machinery above, I estimate the same matched difference-in-differences model, evaluating how wealth moves following a child's disability insurance (DI) take-up. I restrict the events to follow stable employment histories to focus on sudden events, and require the children

FIGURE 6: CHILD LIFE SHOCKS, HETEROGENEITY BY NUMBER OF CHILDREN

(a) Death of a child



(b) Birth of a grandchild



Notes. Matched difference-in-differences estimates of (12), estimated separately for households with one, two, and three or more children. The horizontal axis is event time, with the event year $k = 0$ normalized to zero. In panel (a) the outcome is household net wealth net of cumulative reverse bequests; in panel (b) it is household net wealth. Vertical bars are 95 percent confidence intervals from standard errors clustered on the matched pair. Appendix Table C4 reports the corresponding estimates.

to be at least two years in the DI scheme so that I evaluate permanent shocks.

In section 5.1, I used a static model to demonstrate how a household allocates resources between own consumption and bequests in old age. In practice, households can also transfer resources while they are alive to smooth shocks to their heirs. For instance, following a grandchild birth, households could increase inter vivo transfers to help offset child-rearing costs. In the disability event, this could be even more pronounced. A household decides between own consumption, transfers today, and bequests tomorrow. Since I only track wealth, it is only possible to have a suggestive answer to which one dominates.

Conceptually, following a decline to child's lifetime earnings capacity, altruistic parents wish to increase transfers and bequests, financed through a decrease in own consumption. A negative effect following the disability event can speak to increased transfers today. A positive effect would imply the opposite: that households increase wealth reserved for intended bequests. On the other hand, a null effect would likely mean that households do not alter their consumption in response to earnings shocks to heirs. [15]

Figure 7 reports the estimation results in panel (a). Following the disability insurance shock, there is a clear downward trend in the wealth of parents compared to control households. The effect is around €-1,300 per year, amounting to around €-6,800 by year five. The findings are in line with altruistic parents supporting their disabled children when they are alive. To get to the precise effect of this on bequests, it would be useful to have a sense of the size of the inter-vivo transfers, to ascertain whether the bequests also increase or stay stable. Panels (b) and (c) report raw trends and heterogeneity by number of children. Based on the raw trends, matching seems to do decently well in the pre-periods, and the effects by number of children largely follow each other.

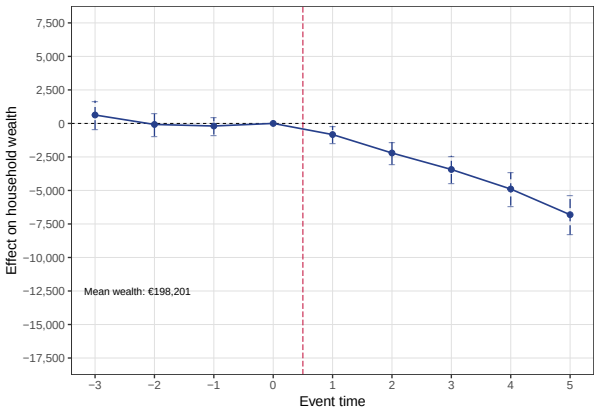
Discussion. Findings documented in this section demonstrate that changes to the set of beneficiaries move household savings decisions in old age. Furthermore, these changes go in line with as predicted under an *intended* bequest motive: decreases in the set of beneficiaries cause a lasting decrease in the wealth holdings, and increases in the set moves wealth upwards. An alternative scenario would have been that households do not care about to whom they bequeathe their wealth, whether that may be the child, another family member, or charity; but merely enjoy the act of giving to someone, or hold wealth purely because they get utility from wealth itself.

Taking these results at face value, it is possible to conduct a simple back-of-the-envelope exercise to quantify this intended bequest effect. I take the magnitude of the last coefficient estimated in the child death and grandchild birth events, divide by the mean wealth in the estimation samples, and multiply by the mean number of children and grandchildren.

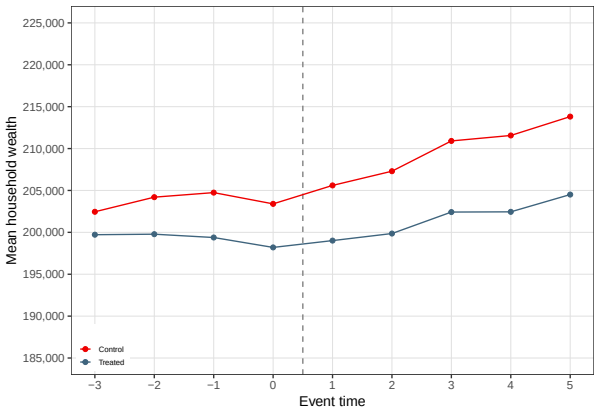
[15] it is also possible that the increased transfers today and increases in wealth reserved for intended bequests offset each other, but this is plausibly less likely.

FIGURE 7: CHILD DISABILITY EVENT RESULTS

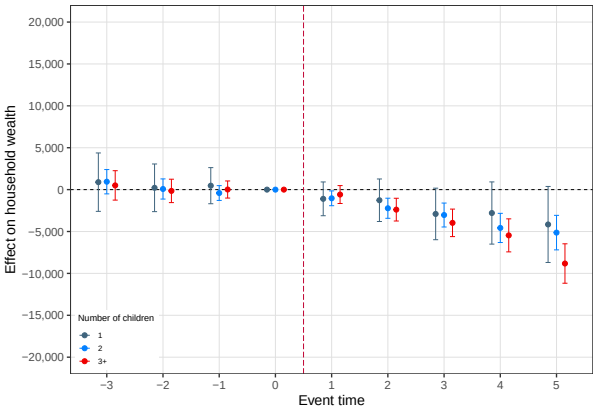
(a) Matched difference-in-differences



(b) Raw wealth trends



(c) Heterogeneity by number of children



Notes. The event is a child's take-up of disability insurance, restricted to children with stable employment histories who remain in the scheme for at least two years. Panel (a) reports matched difference-in-differences estimates of (12); vertical bars are 95 percent confidence intervals from standard errors clustered on the matched pair. Panel (b) reports mean wealth by event time for treated households and their matched controls; confidence intervals are omitted for readability. Panel (c) repeats panel (a) separately for households with one, two, and three or more children. In all panels the outcome is household net wealth, and the event year $k = 0$ is normalized to zero.

$$\begin{aligned}\text{Intended bequests} &= \sum_{i\in\{\text{child,grandchild}\}} \frac{\left|\hat{\beta}^{i}_{k=5}\right|}{\hat{\mu}^{i}_{wealth}} \times N_i \\ &= \frac{€11{,}630}{€210{,}142} \times 2.53 \;+\; \frac{€3{,}328}{€289{,}547} \times 3.05 \\ &= 0.140 + 0.035 \;=\; 0.175\end{aligned}$$

The exercise attributes roughly 17.5 percent of household wealth to an intended bequest motive, of which 14.0 percentage points come from the child margin and 3.5 from the grandchild margin. Table 6 reports the back-of-the-envelope results. Averaging the effect over the five post-event years instead gives 13.2 percent. All in all, intended bequests seem to have a large effect on

TABLE 6: BACK-OF-THE-ENVELOPE INTENDED BEQUEST SHARE

	Death of a child		Birth of a grandchild		Total
	$\hat{\beta}_i$	Implied share of wealth (%)	$\hat{\beta}_i$	Implied share of wealth (%)	Implied share of wealth (%)
	(1)	(2)	(3)	(4)	(5)
Effect at $k = 5$	−11,630	14.0	3,328	3.5	17.5
	(1,756)	(2.1)	(1,597)	(1.7)	(2.7)
Average post-event effect, $k = 1, \ldots, 5$	−8,816	10.6	2,491	2.6	13.2
	(1,289)	(1.5)	(1,182)	(1.2)	(2.0)
Trend only, extrapolated to mean age at death	−6,060	7.3	3,230	3.4	10.7
Mean wealth, $\hat{\mu}_i$	€210,142		€289,547		
Mean number of beneficiaries, N_i	2.53		3.05		

Notes. Columns (2) and (4) divide $|\hat{\beta}_i|$ by mean wealth $\hat{\mu}_i$ and multiply by the mean number of beneficiaries N_i; column (5) sums them. Standard errors for the effect at $k = 5$ are those of the coefficient; for the average post-event effect they are the mean of the post-event standard errors, an upper bound on the true standard error. The last row extrapolates the post-event trend linearly to the expected age at death: the slope of the coefficients over $k = 1, \ldots, 5$, net of the pre-event slope over $k = -2, \ldots, 0$ ($k = -3$ is the matching anchor), accumulated from zero at $k = 1$ to 12.8 years (child death) and 15.4 years (grandchild birth) after the event. These horizons are the expected remaining lifetime at the sample's mean household age at the event (Table C2) under the Dutch period life table for 2023.

the wealth holdings in old age, comparable to the precautionary savings effect calculated in section 4.

There are some obvious caveats due to the simplicity of the estimation. First, the multiplication step assumes that the household puts aside a per-child earmark b, which is constant in N. The alternative would've been that a fixed estate B is split in N ways. There is some support for this in the data: heterogenous results show similar patterns for households with two and more than three children. In particular, the per child earmark predicts $\beta_5^2 = \beta_5^{3+}$, and the observed difference is -745 (se 2,708). On the other hand, the fixed estate split N ways would predict $2 \times \beta^2 = 3 \times \beta^{3+}$, which does not find support in the data.

Second, the effect is assumed to have converged at the end of the estimation window. This seems true for grandchild births, but there is a clear negative slope on the child death results that might lead to underestimation of the effect. The last row of Table 6 relaxes this assumption: extrapolating only the linear trend from $k = 1$ to $k = 5$ to the expected age at death yields 10.7 percent, without the sharp drop in the first year. Third, the BOE assumes all of the β is attributable to the bequest motive, i.e. that exclusion restriction holds. Finally, it is worth noting that here there is no double counting since I focus on deaths of children without heirs.

6 Conclusion

In this paper, I study why retirees draw down so little wealth, a puzzling empirical pattern. Despite the generous welfare state, I estimate that the average Dutch retiree decumulates less than one-fifth of their private wealth over two decades, and above 40% die with at least as much real wealth as they entered retirement with. I find that precautionary motives against long-term-care expenditures in combination with the bequest motive can explain a portion of this pattern. I estimate that old age households accumulate €0.45 of wealth in anticipation of a €1 increase in long-term-care costs. This marginal response translates into up to 22% of wealth attributable to expected care payments. In order to estimate the effect of bequest motives on wealth, I exploit life shocks to the set of beneficiaries of a household. Changes in the number of beneficiaries change savings behavior in a way that aligns well with an *intended* bequest motive towards heirs. Losing a child beneficiary is linked to a decrease of 5.5% of mean wealth, while gaining an unexpected grandchild leads to a small but significant increase of around 1.2% five years following the event. Together, the two key motives account for up to 40% of wealth held late in life.

The findings shed light on the savings behavior of retirees. First, they show that intended bequests directed towards beneficiaries affect savings decisions. This supports the presence of an intentional bequest motive as opposed to accidental bequests arising from premature deaths or a bequest motive that is indifferent to the beneficiary identity. Second, the results show that both the precautionary motives against care expenses, and bequest motives together are important in explaining the absence of dissaving. Extrapolations of the estimated coefficients from reduced-form designs put the effects close to one another. These results imply that decreasing care copayments would lower the private saving of households. Furthermore, the evidence on the recipient-specific bequest motive documented in this paper suggests that heir-specific inheritance tax rates would have different welfare consequences than a tax on the total estate, as households respond to marginal changes in the set of beneficiaries.

Finally, the two key motives cannot account for the puzzle fully. According to my estimates, the precautionary and bequest motives leave about 60 percent of wealth held late in life unexplained. While outside the scope of this paper, several other explanations could close this gap. One set of explanations relates to whether households treat different assets as fungible. If households finance consumption out of income flows such as pensions but treat accumulated wealth as principal not to be drawn down, as in models of mental accounting, wealth would remain flat even in the absence of the key motives. Comparisons of consumption following windfalls, such as inheritances received, with how it responds to changes in pension income can help test this mechanism. A second set of potential explanations relates to habits and inattention. Consumption patterns formed during working life may persist into retirement, which would keep spending below the level allowed by private wealth. Third, the marginal utility of consumption may fall as health deteriorates (Finkelstein et al., 2013), and cognitive decline can make financial planning harder in old age (Mazzonna and Peracchi, 2024). Both would slow the drawdown of wealth at older ages. Lastly,

despite the slow drawdown of wealth excluding home equity, the interaction of home ownership with bequest and precautionary motives, as well as housing as a consumption flow itself, deserves more attention (Nakajima and Telyukova, 2020; French et al., 2023). Distinguishing among these channels is a fruitful direction for future work.

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Appendix A: Wealth Profiles in Old Age

TABLE A1: SUMMARY STATISTICS AT SAMPLE ENTRY

	Mean	Std. Dev.	P10	P25	P50	P75	P90
Panel A. All households							
Average household age	70.1	5.9	65.0	65.0	67.4	74.0	79.5
Share with any child	0.71						
Number of children	1.89	1.72	0.00	0.00	2.00	3.00	4.00
Net wealth	270,000	2,518,721	1,026	10,651	91,411	314,750	606,475
Net wealth excl. home equity	144,722	2,491,682	670	6,215	26,697	91,106	280,584
Financial wealth	95,259	2,226,067	1,074	6,416	26,030	75,759	207,394
Pension income	25,489	21,308	1,629	14,027	21,847	32,926	48,622
Number of households	3,111,112						
Panel B. Singles							
Average household age	70.7	6.5	65.0	65.0	67.5	76.0	81.5
Share with any child	0.55						
Number of children	1.47	1.78	0.00	0.00	1.00	2.00	4.00
Net wealth	196,348	893,327	42	4,704	26,840	236,648	501,435
Net wealth excl. home equity	105,487	837,988	0	3,473	19,655	67,100	223,645
Financial wealth	78,365	333,288	469	3,698	19,065	58,278	179,698
Pension income	16,740	15,156	1	5,705	15,283	22,111	32,820
Number of households	1,754,796						
Panel C. Couples							
Average household age	69.3	4.8	65.0	66.0	67.0	71.5	76.5
Share with any child	0.93						
Number of children	2.45	1.46	1.00	2.00	2.00	3.00	4.00
Net wealth	345,301	3,446,150	3,918	26,632	179,770	381,939	705,618
Net wealth excl. home equity	183,886	3,421,661	1,839	11,387	30,782	116,515	343,949
Financial wealth	113,297	3,150,200	2,487	11,240	28,454	93,474	236,748
Pension income	37,299	22,788	19,662	24,091	31,255	44,104	61,454
Number of households	1,304,602						

Notes. Summary statistics for the main sample of Table 1, measured in the first year each household is observed. Monetary values are in real terms, in 2015 euros (deflated with the consumer price index). Net wealth is total private wealth, excluding annuities; financial wealth is the value of financial assets. Pension income is gross annual pension income, summed over household members. Couples comprise married or cohabiting households and widowed households, i.e. including the household-years after the death of a spouse; the full sample in Panel A also includes divorced households. For confidentiality, each percentile is reported as the mean of the observations within one percentile of the quantile.

A.1 Life-cycle model

Setup. A household alive at age $a_0 = 67$ chooses a consumption path $\{c_a\}$ to maximize expected discounted lifetime utility, weighting each future period by the probability of surviving to enjoy it:

$$\max_{\{c_a\}} \mathbb{E} \sum_{a=a_0}^{A} \beta^{\,a-a_0}\, s_a\, u(c_a), \qquad u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \tag{A.1}$$

where $s_a = \Pr(\text{alive at } a \mid \text{alive at } a_0)$ is unconditional survival ($s_{a_0} = 1$), β the discount factor, γ the coefficient of relative risk aversion, and $A = 110$ the maximum survivable age. Wealth evolves deterministically,

$$w_{a+1} = (1+r)\,(w_a + y - c_a), \qquad w_{a+1} \geq 0, \qquad w_{a_0} = W_0, \tag{A.2}$$

with real net return r and a constant real pension income y, received at the start of each year alive and financed outside private wealth. Private wealth cannot be borrowed against future pension income. There is *no bequest motive*—wealth left at death is purely accidental—and *no annuity market*, so the agent cannot trade away longevity risk. Throughout I impose the discipline device $\beta(1+r) = 1$, which nets out the pure time-preference/interest tilt so that mortality is the only force shaping the consumption path.

Euler equation. The first-order condition, written with the one-year conditional survival probability $p_a \equiv s_{a+1}/s_a$ and $u'(c) = c^{-\gamma}$, is

$$\frac{c_{a+1}}{c_a} = \big(\beta(1+r)\,p_a\big)^{1/\gamma}. \tag{A.3}$$

Lower survival ($p_a < 1$) front-loads consumption ("I might be dead next year"); higher risk aversion γ dampens that front-loading. Cumulating (A.3) from a_0,

$$c_a = c_{a_0}\, G_a, \qquad G_a = \prod_{k=a_0}^{a-1} \big(\beta(1+r)\,p_k\big)^{1/\gamma}, \quad G_{a_0} = 1. \tag{A.4}$$

The budget that pins down c_{a_0}. Because wealth dynamics are deterministic and there is no annuity market, the agent must remain solvent even in the state where they survive to the maximum age A. Iterating (A.2) forward with the terminal condition $w_{A+1} = 0$ yields the intertemporal budget

$$\sum_{a=a_0}^{A} \frac{c_a}{(1+r)^{a-a_0}} = W_0 + y \underbrace{\sum_{a=a_0}^{A} (1+r)^{-(a-a_0)}}_{\equiv\ \Lambda}, \tag{A.5}$$

and substituting (A.4) and solving,

$$c_{a_0} = \frac{W_0 + y\,\Lambda}{\Phi}, \qquad \Phi \equiv \sum_{a=a_0}^{A} G_a\,(1+r)^{-(a-a_0)}. \tag{A.6}$$

Equation (A.6) is the unconstrained solution. With $y > 0$, the path it implies falls below zero wealth at old ages, as the agent would like to borrow against future pension income. Under $\beta(1+r) = 1$, once the no-borrowing constraint binds it binds in every later year, so the constrained solution follows (A.3) until private wealth is exhausted at some age τ, with the sums in (A.5) and in Φ running to $\tau - 1$, and consumes the pension, $c_a = y$, thereafter. I solve for τ numerically by comparing expected utility across all candidate ages. With $y = 0$ the constraint never binds and (A.6) is the solution. Crucially, the budget (A.5) is *not* survival-weighted: survival enters the objective—and hence the consumption tilt G_a—but not the resource constraint, because without annuities the agent keeps their wealth whether or not they die. This is the formal origin of the retained wealth. (With actuarially fair annuities the budget would be survival-weighted, wealth would be forfeited at death, and the model would collapse to the deterministic full-decumulation case.)

Deterministic special case. Setting $p_a = 1$ for $a < T$ and $p_a = 0$ afterwards, with T the known death age, (A.3) reduces to $c_{a+1}/c_a = (\beta(1+r))^{1/\gamma}$, which under $\beta(1+r) = 1$ equals one for *every* γ: consumption is flat and the entire path is γ-invariant. With no bequest the agent consumes their last euro at T, so decumulation is complete. With no mortality risk, risk aversion does not affect decumulation at all.

Comparative statics. Two analytical predictions motivate the robustness rows in Table A2. Both are derived on the unconstrained path; the rows of the table are the constrained solutions. Write the proportional first-year dissaving rate, from (A.2), as

$$\delta \equiv \frac{W_0 - w_{a_0+1}}{W_0} = \frac{(1+r)\,(c_{a_0} - y) - rW_0}{W_0}. \tag{A.7}$$

(1) Adding pension income. From (A.6), on the unconstrained path, $\partial c_{a_0}/\partial y = \Lambda/\Phi > 0$, so

$$\frac{\partial \delta}{\partial y} = \frac{(1+r)\,(\Lambda/\Phi - 1)}{W_0} \;>\; 0, \tag{A.8}$$

since $\Phi = \sum_a G_a(1+r)^{-(a-a_0)} < \sum_a (1+r)^{-(a-a_0)} = \Lambda$ (under longevity risk $G_a < 1$ for $a > a_0$). A larger pension unambiguously *raises* private-wealth dissaving: guaranteed lifelong income substitutes for the precautionary wealth that longevity risk had made the agent hold, so private wealth is drawn down faster. The no-borrowing constraint weakens but does not reverse this effect: in the baseline calibration, first-year dissaving rises from 2.4% without pension income to 3.5% with it, against 4.4% on the unconstrained path. Conversely, removing pension income, the "No pension income" row of Table A2, slows decumulation.

(2) Increasing the mean age at death. An outward shift of the survival curve raises the conditional survival probabilities $\{p_k\}$. From (A.4), $\partial G_a/\partial p_k = G_a/(\gamma\, p_k) > 0$ for each $k < a$, so Φ rises and,

since Λ does not depend on survival, $c_{a_0} = (W_0 + y\Lambda)/\Phi$ falls on the unconstrained path:

$$\frac{\partial c_{a_0}}{\partial p_k} = -\frac{W_0 + y\Lambda}{\Phi^2}\frac{\partial \Phi}{\partial p_k} < 0, \qquad \frac{\partial \delta}{\partial p_k} < 0. \tag{A.9}$$

A longer expected life *lowers* initial consumption and *slows* decumulation: the agent stretches wealth over more years and holds a larger buffer against the right tail. In the constrained solution reported in the "90th-percentile (long-lived) survival" row, the annual decumulation rate falls from 10.3 to 7.8%/yr.

A.2 Calibration

Baseline parameters: $a_0 = 67$, $A = 110$, $r = 0.03$, $\beta = 1/(1+r) \approx 0.971$, $\gamma = 4$, $y = 0.067$, and $W_0 = 1$ (paths are shares of wealth at retirement entry, so y is annual pension income as a share of initial wealth). Mortality is taken directly from the Dutch period life table (Statistics Netherlands, 2023, via the Human Life-Table Database, sex-averaged): one-year survival is $p_a = 1 - q_a$ for ages 67–110, where q_a is the empirical age-specific death probability. The published table closes near age 99; I extend the tail to $A = 110$ by continuing the upper-age log-hazard (a Gompertz-shaped extrapolation) rather than imposing the table's terminal death wall. This baseline ("mean") calibration implies a remaining life expectancy $e(67) = 18.6$ years (mean age at death 86). The long-lived calibration scales the age-specific hazard by a constant factor to hit $e(67) = 28$ years (mean age at death 95), preserving the empirical age-shape of mortality. A Gompertz hazard $\mu(a) = \exp(\alpha + \rho a)$ with slope $\rho = 0.11$, calibrated to life-expectancy targets of 18.5 and 28 years, reproduces the annual decumulation rates to within about 0.6 percentage points and is retained only as a cross-check; its drawdown shares are up to 5 percentage points lower, partly because its mean age at death rounds to 85. Table A2 reports the resulting decumulation across risk aversion, the absence of pension income, and the long-lived calibration.

TABLE A2: LIFE-CYCLE BENCHMARK: ROBUSTNESS

	Age at death	Annual decum. rate (%/yr)	Drawn down by death (%)
Baseline			
Uncertain lifetime ($\gamma = 4$, pension income)	86	10.3	78.1
Alternative risk aversion			
$\gamma = 2$	86	15.4	90.7
$\gamma = 5$	86	9.0	73.9
$\gamma = 10$	86	6.1	61.1
Baseline ($\gamma = 4$) with			
No pension income	86	5.1	57.0
90th-pct. (long-lived) survival	95	7.8	82.8

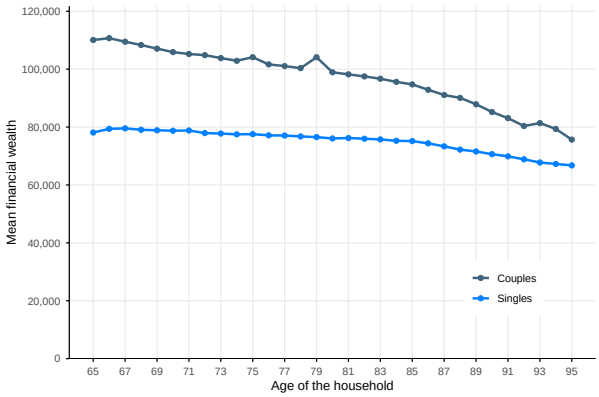
Notes. Every row is the uncertain-lifetime (longevity-risk) benchmark of Table 2, with $r = 3\%$ and $\beta = 1/(1+r)$. The three panels report, respectively, the $\gamma = 4$ baseline with pension income; the same model under alternative risk aversion, keeping the pension; and the baseline with one feature changed at a time. The baseline pension is a constant real income of 6.7% of initial wealth per year, received at the start of each year, financed outside private wealth and solved under a no-borrowing constraint. The features in the third panel are each applied to the $\gamma = 4$ baseline (they are *not* cumulative with the risk-aversion rows above): (i) no pension income; and (ii) a "long-lived" calibration with remaining life expectancy of 28 years at 67 (mean age at death 95, the 90th percentile of the baseline distribution of age at death), a longevity stress test. Both metrics use the same definitions as Table 2: the *annual decumulation rate* is the survival-weighted slope of log wealth on age over the years in which private wealth is positive, and *drawn down by death* is the end-of-year share of age-67 wealth consumed by the (row-specific) mean age at death. Removing pension income or raising risk aversion slows private-wealth drawdown, and a longer horizon lowers the annual rate—yet none approaches the observed drawdown (16% by the mean age at death).

A.3 Additional Wealth-Age Profiles

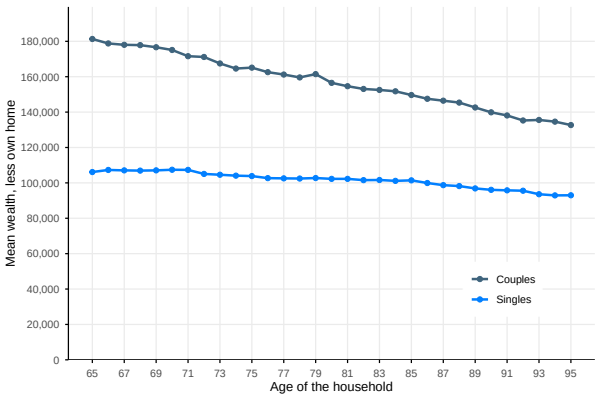
This subsection reports the empirical wealth-age profiles that underlie the flat mean profile in Figure 1. Figure A1 shows that the pattern is not driven by home equity: both liquid financial wealth and net wealth excluding housing remain broadly flat in old age. Figure A2 disaggregates the profile by birth cohort and by age at death, and Figure A3 reports the median (rather than mean) profile by birth cohort. All panels use the same Dutch administrative population data as Figure 1.

FIGURE A1: MEAN WEALTH BY AGE: ALTERNATIVE WEALTH CONCEPTS

(a) Financial wealth



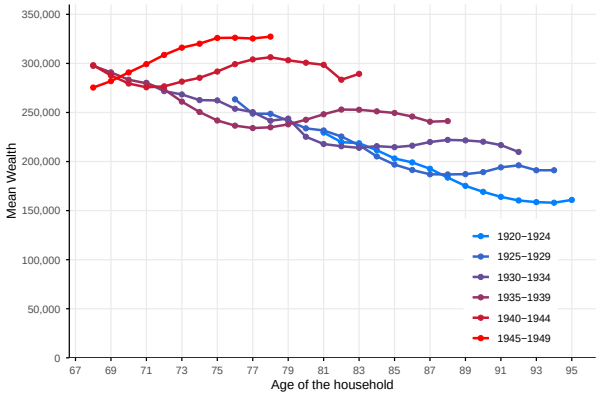
(b) Wealth, excluding home equity



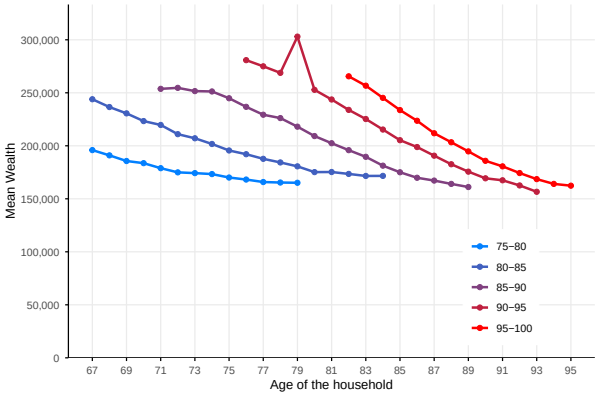
Notes. Mean wealth by single year of age, household fixed-effects specification. Panel (a) uses liquid financial wealth; panel (b) uses net wealth excluding home equity.

FIGURE A2: WEALTH BY AGE, DISAGGREGATED BY BIRTH COHORT AND AGE AT DEATH

(a) By birth cohort

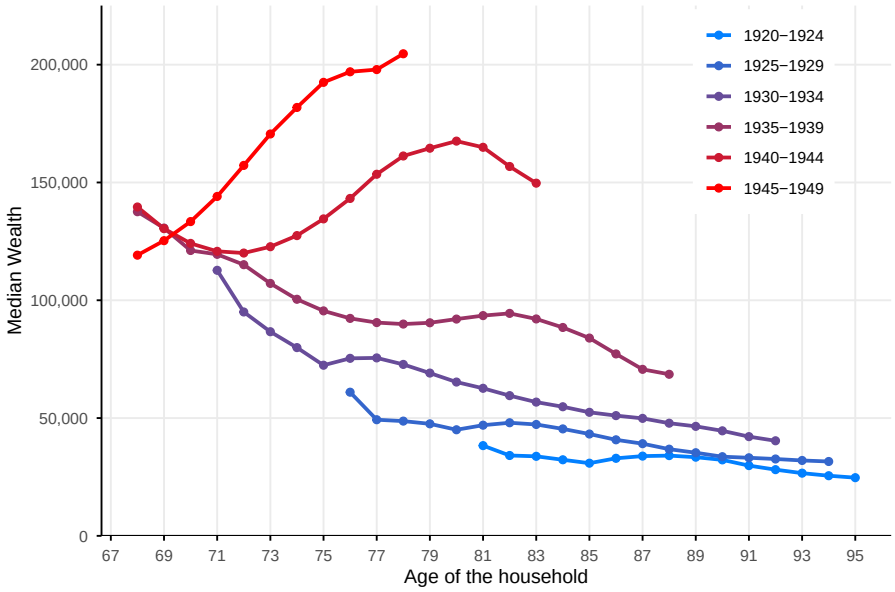


(b) By age at death



Notes. Raw mean net wealth by single year of age. Panel (a) groups households by birth cohort; panel (b) groups households by age at death.

FIGURE A3: MEDIAN WEALTH BY AGE AND BIRTH COHORT



Notes. Median net wealth by single year of age, by birth cohort. Dutch administrative population data.

TABLE A3: IMPLIED WEALTH DECUMULATION BY ESTIMATION SAMPLE

	Entry sample	Death sample	Full sample
Pooled	10.0%	34.7%	15.9%
Singles	−3.4%	28.2%	11.8%
Couples	17.4%	46.1%	23.5%

Notes. Implied cumulative wealth decumulation from the household fixed-effects model in equation (1), estimated under a cubic age profile and evaluated at the mean age at death. Columns report three estimation samples: the *entry sample* (households first observed at ages 67–70, right-censored), the *death sample* (households observed to die within the panel, left-truncated), and the *full sample*. The entry and death samples bracket the full-sample decumulation both in the pooled sample and within each marital-status group. Dutch administrative population data.

A.4 Selection-Correction Weighting for Realized Drawdown

This appendix details the weighting behind Panel A of Table 3. The realized drawdown rate $D_i = 1 - W_i^{\text{death}}/W_i^{\text{initial}}$ is defined only for *completed spells*—households observed both near retirement entry (anchor wealth at ages 67–70) and near death. Two forms of selection separate the completed-spell median from the population median: whether a spell is observed at both endpoints is not random, and the age-at-death composition of completed spells differs from that of the full decedent population. I correct these in sequence.

Censoring model. I restrict attention to the cohort window of birth years 1936–1948, for which both an anchor and a death observation can in principle fall inside the 2006–2024 panel. For every household i in this window with non-missing anchor wealth, let $R_i = 1$ if it is observed at both endpoints—a completed spell—and $R_i = 0$ otherwise. I model the probability of being observed at both endpoints with a logit on predetermined covariates,

$$\Pr(R_i = 1 \mid X_i) = \Lambda\Big(\theta_0 + \theta_1 \, W_i^{\text{initial}} + \theta_2 \, \text{Couple}_i + \theta_3 \, \text{Birthyear}_i\Big), \tag{A.10}$$

where $\Lambda(\cdot)$ is the logistic CDF, W_i^{initial} is winsorized anchor wealth, Couple_i indicates a two-person household, and Birthyear_i is the main earner's birth year. The parameters θ are estimated by maximum likelihood on *all* cohort-window households, observed and censored alike.

Inverse-probability weights. Each completed spell receives the inverse of its estimated probability of being observed at both endpoints,

$$\omega_i^{\text{IPCW}} = \frac{1}{\widehat{\Pr}(R_i = 1 \mid X_i)}. \tag{A.11}$$

A household that resembles those we typically fail to observe at both endpoints—low predicted $\widehat{\Pr}(R_i = 1 \mid X_i)$—receives a large weight, so the weighted completed-spell sample recovers the covariate distribution of the full at-risk population (Robins and Finkelstein, 2000).

Age-at-death weights. I sort households into age-at-death bins $a \in \{67\text{–}74,\ 75\text{–}79,\ 80\text{–}84,\ 85\text{–}89,\ 90+\}$. Let π_a denote bin a's share among *all* decedents in the data (the population target) and q_a its share among completed spells in the cohort window (the sample). The age-at-death correction ω_a replaces the sample shares q_a with the population shares π_a when the bins are combined, so that the estimate reflects the population's age-at-death mix rather than the sample's.

Weighted median. For a set of drawdown rates $\{D_i\}$ with nonnegative weights $\{\omega_i\}$, the weighted median is the value at which cumulative weight first reaches one half: sort the D_i in ascending order, form the normalized cumulative weights $\sum_{j \le k} \omega_j / \sum_j \omega_j$, and take the smallest D_i at which this sum is at least $1/2$. With IPCW weights, the median is thus the point that splits the *reweighted*, population-representative mass in half rather than the observed sample mass.

Combining the corrections. The corrected statistics are computed in two stages. Within each age-at-death bin a, I compute the IPCW-weighted median drawdown m_a using the weights in (A.11). The bin medians are then aggregated by target age-at-death shares,

$$\widehat{D} = \sum_a \lambda_a \, m_a. \tag{A.12}$$

Setting $\lambda_a = q_a$ (sample shares) yields the *IPCW* row: censoring is corrected within bins, but the age-at-death mix is left at its sample value. Setting $\lambda_a = \pi_a$ (population shares) yields the *IPCW* $+ \omega_a$ row, which additionally standardizes the age-at-death composition to the population; the move from the first to the second is exactly the ω_a correction. The naive row applies neither and is the unweighted pooled median of D_i over completed spells.

Interpretation. Because birth year enters the censoring model (A.10) and, within the fixed calendar window, birth year bounds the age at which death can be observed, the IPCW step already shifts the sample's age-at-death composition part of the way toward the population. The decomposition into an IPCW step and an ω_a step is therefore sequential rather than orthogonal: their sum recovers the full correction, but the individual pieces should not be read as independent contributions.

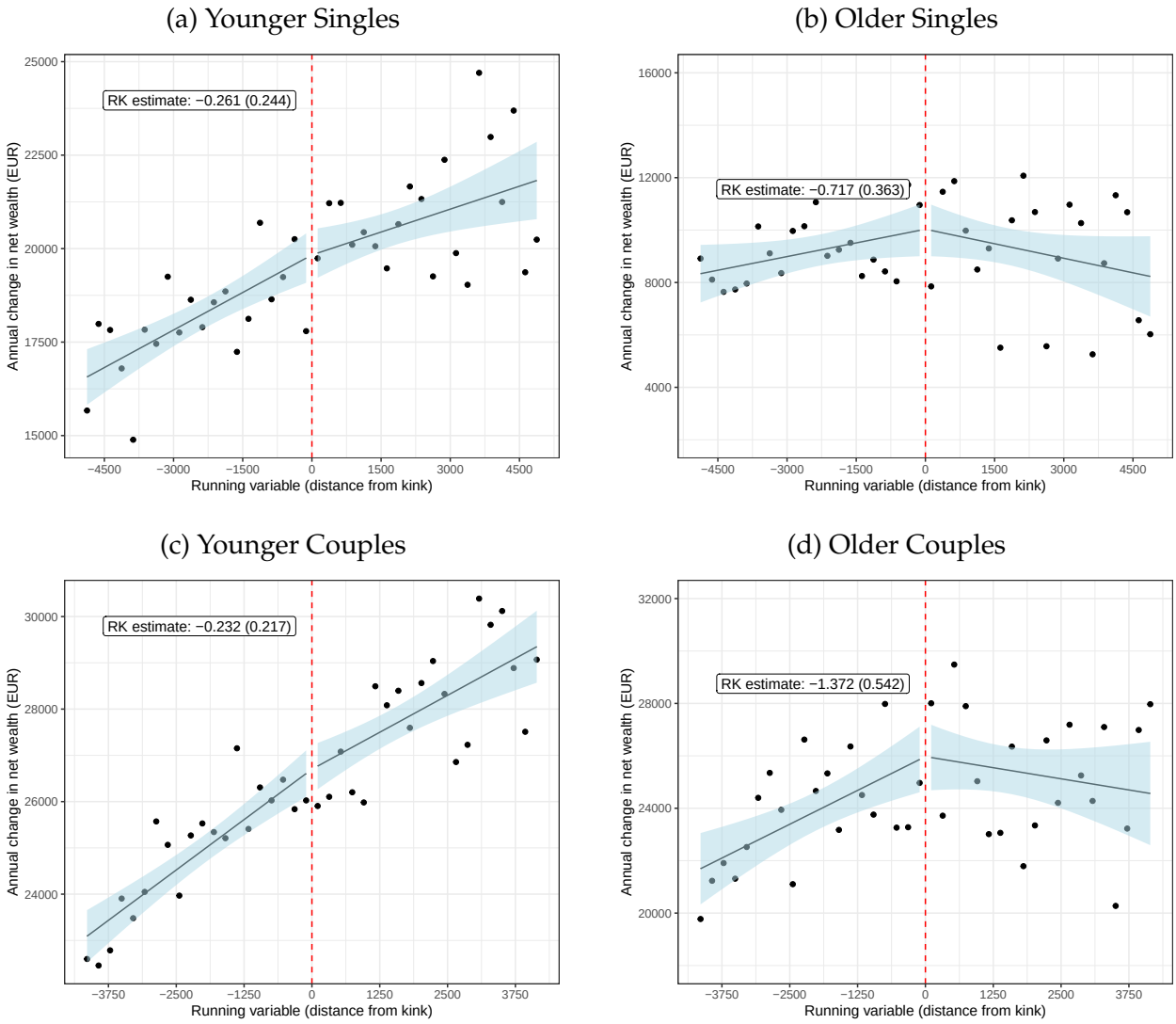
Appendix B: Precautionary Savings against Care Expenditures

TABLE B1: SUMMARY STATISTICS: REGRESSION KINK SAMPLES

	Mean	Std. Dev.	P10	P25	P50	P75	P90
Panel A. Singles							
Average household age	77.4	7.0	69.6	72.0	76.0	82.5	87.5
Share with any child	0.65						
Number of children	1.55	1.49	0.00	0.00	2.00	2.00	3.00
Net wealth	319,645	297,616	25,144	120,552	277,324	457,191	646,292
Net wealth excl. home equity	128,956	190,374	8,100	36,982	97,701	181,076	276,652
Wealth percentile among all households	71.3						
Number of households	127,709						
Number of observations	353,352						
Panel B. Couples							
Average household age	75.3	4.9	69.6	72.0	74.5	78.0	82.5
Share with any child	0.94						
Number of children	2.29	1.18	1.00	2.00	2.00	3.00	3.24
Net wealth	359,590	307,377	61,694	185,145	326,827	479,796	654,781
Net wealth excl. home equity	105,568	207,157	9,830	30,144	68,101	134,671	224,603
Wealth percentile among all households	58.8						
Number of households	187,369						
Number of observations	616,131						

Notes. Summary statistics for the regression kink estimation samples: household-years within €5,000 (singles) and €4,250 (couples) of the copayment cap in terms of the running variable, 2017–2023. Singles are single households and couples are married or cohabiting households; the reference person has no wage or self-employment income. Monetary values are in nominal euros. Net wealth is total private wealth, excluding annuities. Wealth percentile is the household's percentile rank (0–100) in the net wealth distribution of all households of the same type (single or couple) in which all members are aged 65 or older, in the same year. Among all household-years that satisfy the sample restrictions in 2017–2023, the contribution base exceeds the cap in 13% of single and 25% of couple household-years. For confidentiality, each percentile is reported as the mean of the observations within one percentile of the quantile.

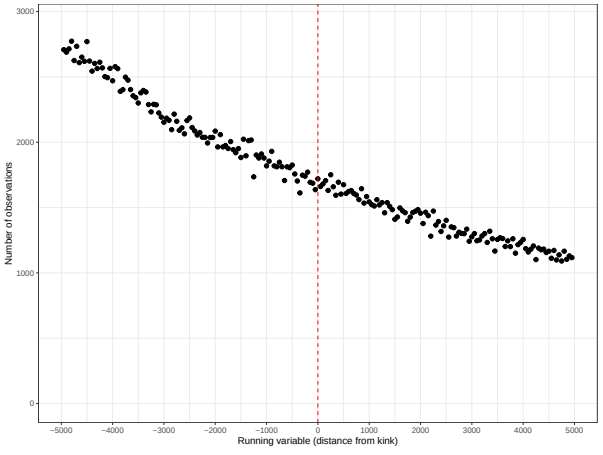
FIGURE B1: SAVINGS RESPONSES TO CARE COSTS



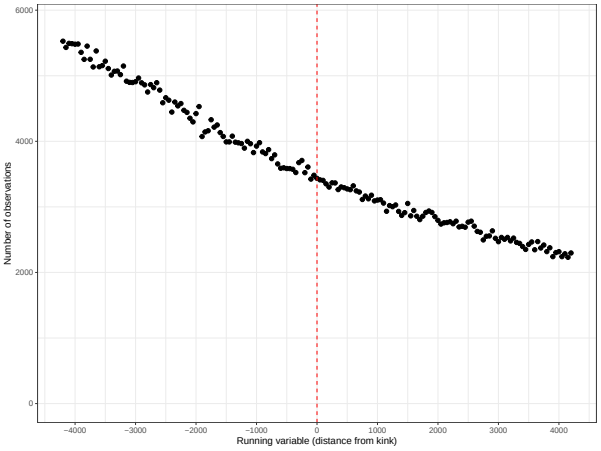
Notes. Regression kink design estimates by household type and age group. Panels (a) and (b) show single-person households below and above age 80; panels (c) and (d) show couple households below and above age 80. The horizontal axis plots the running variable $m_{it} - \overline{C}$, the contribution base centered at the copayment cap. The vertical axis plots wealth accumulation ΔW_{it}. The solid lines are local linear fits on either side of the kink.

FIGURE B2: DENSITY OF RKD SAMPLES

(a) Singles



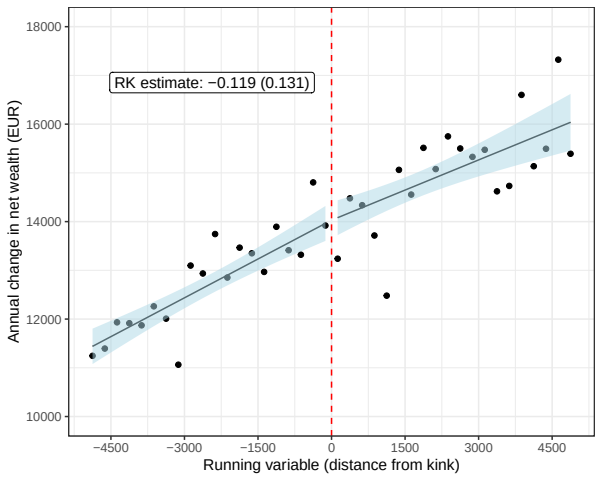
(b) Couples



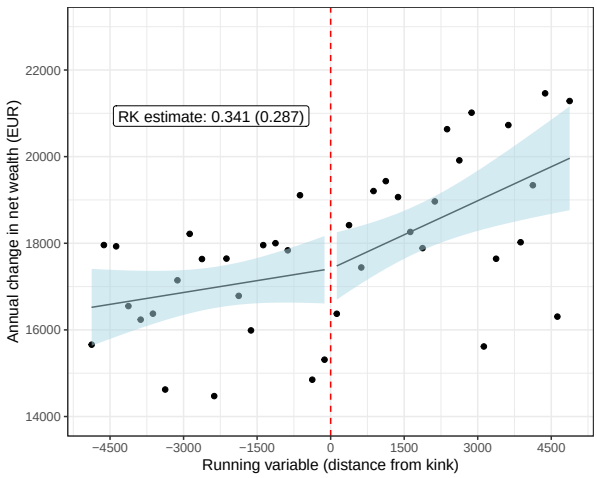
Notes. Density of the running variable $m_{it} - \overline{C}$ around the copayment cap for single-person households (panel a) and couple households (panel b).

FIGURE B3: REGRESSION KINK PLACEBO CHECKS — SINGLES

(a) $\overline{C} - 5000$ Placebo Cutoff



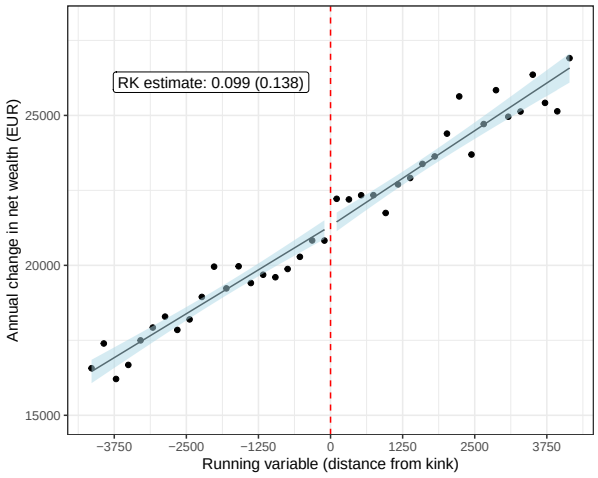
(b) $\overline{C} + 5000$ Placebo Cutoff



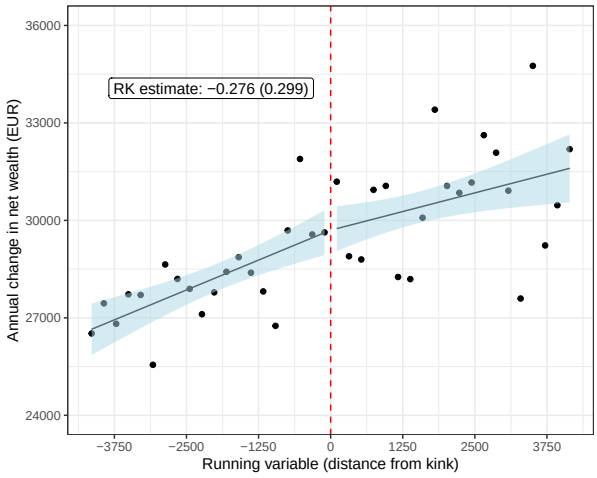
Notes. Placebo regression kink design estimates for single-person households. Panel (a) places the kink at $\overline{C} - 5{,}000$; panel (b) at $\overline{C} + 5{,}000$. Neither placebo cutoff produces a significant slope change. In each panel the running variable is re-centered at the placebo cutoff, so the horizontal axis measures the distance from that cutoff.

FIGURE B4: REGRESSION KINK PLACEBO CHECKS — COUPLES

(a) $\overline{C} - 5000$ Placebo Cutoff



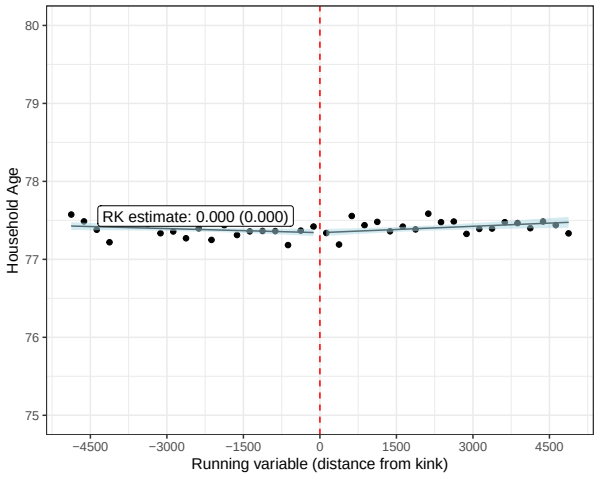
(b) $\overline{C} + 5000$ Placebo Cutoff



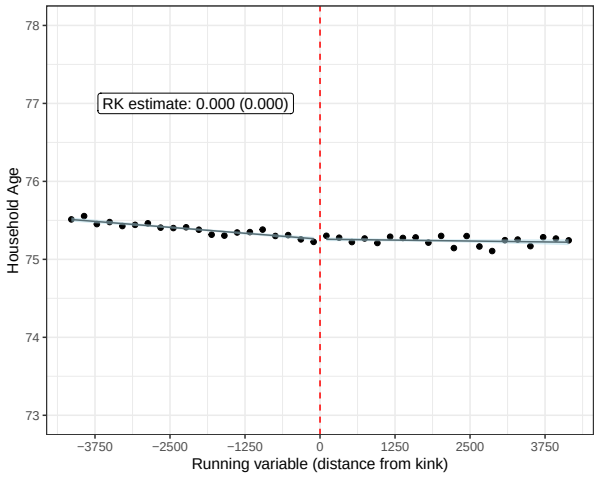
Notes. Placebo regression kink design estimates for couple households. Panel (a) places the kink at $\overline{C} - 5{,}000$; panel (b) at $\overline{C} + 5{,}000$. Neither placebo cutoff produces a significant slope change. In each panel the running variable is re-centered at the placebo cutoff, so the horizontal axis measures the distance from that cutoff.

FIGURE B5: COVARIATE BALANCING: AGE

(a) Singles



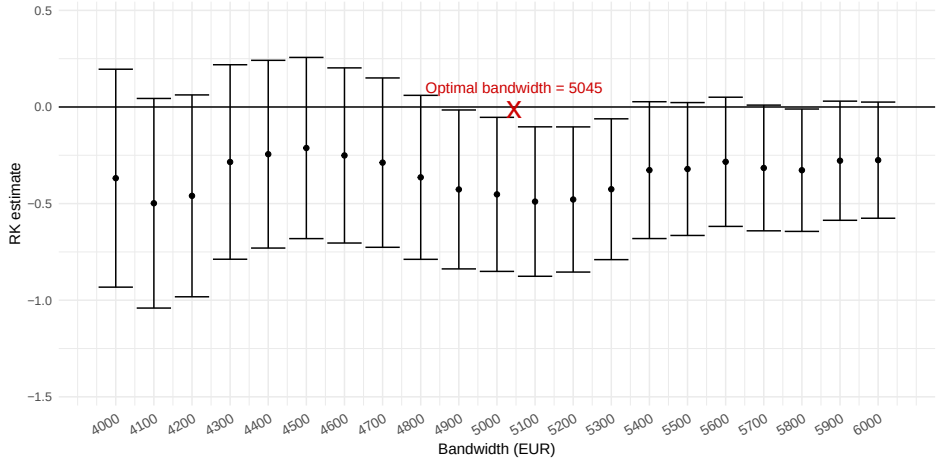
(b) Couples



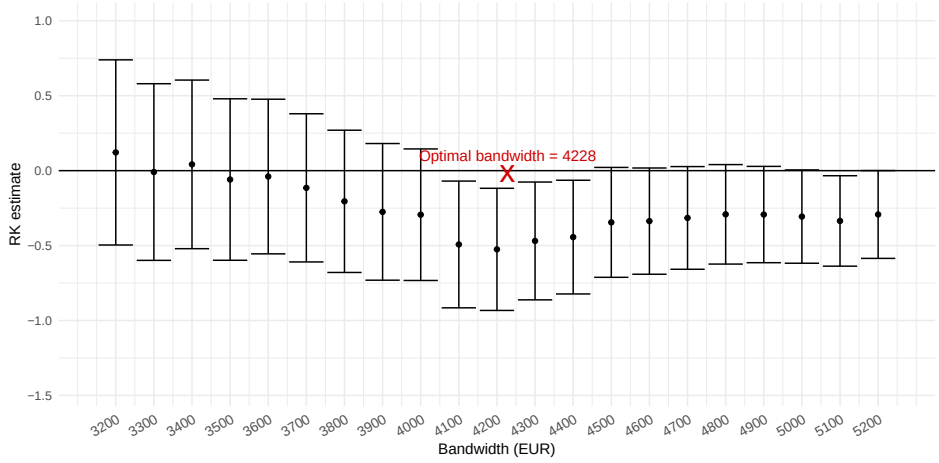
Notes. Covariate balancing test for age. Each panel plots household age against the running variable $m_{it} - \overline{C}$. Slope change of 0 at the kink indicates that age evolves smoothly through the threshold.

FIGURE B6: REGRESSION KINK ESTIMATES BY BANDWIDTH

(a) Singles



(b) Couples



Notes. RKD slope-change estimates as a function of bandwidth for single-person households (panel a) and couple households (panel b).

TABLE B2: CCT OPTIMAL BANDWIDTH RESULTS

	Singles			Couples		
	(1)	(2)	(3)	(4)	(5)	(6)
	Conventional	Bias-Corrected	Robust	Conventional	Bias-Corrected	Robust
Estimate	−0.505	−0.461	−0.461	−0.461	−0.319	−0.319
	(0.205)	(0.205)	(0.307)	(0.212)	(0.212)	(0.304)
Bandwidth	5,045.8	5,045.8	5,045.8	4,228.6	4,228.6	4,228.6
N	356,981	356,981	356,981	612,727	612,727	612,727

Notes. CCT optimal bandwidth estimates for the regression kink design. Standard errors in parentheses.

TABLE B3: BACK-OF-THE-ENVELOPE PRECAUTIONARY WEALTH STOCK: INPUTS

	Singles	Couples
Marginal savings response, $\hat{\beta}_{RKD}$	0.452	0.458
Effective annual copayment	€26,785	€26,785
Expected discounted years of saving	5.879	6.224
Mean age at start	77.4	75.3
Saving ends at	care entry (85) or death	care entry (85) or first death
Precautionary wealth stock	€71,200	€76,400
Mean net wealth	€319,645	€359,590
Share of mean net wealth	22%	21%

Notes. Inputs to Table 4 and equation (7). The precautionary wealth stock is the product of the first three rows, and the share divides it by mean net wealth. The marginal savings response is the RKD estimate of equation (6). The effective annual copayment is the duration-weighted annual copayment under the 2021 schedule: the low rate (€10,000 per year) for the first four months in care and the high rate (€29,630, the annual cap) thereafter, over a mean stay of 2.3 years (Alders et al., 2025). The expected discounted years of saving integrate, from the mean age of each household type to the mean age at nursing home admission, 85 (Alders et al., 2025), the probability that the household is still saving, discounted at a real return of 3%, the same return as the life-cycle benchmark. The mean age is the mean in the headline RKD sample (Table B1). Survival is from the Dutch period life table (Statistics Netherlands, 2023, via the Human Life-Table Database, averaged over men and women); for couples it is the probability that both spouses are alive.

Appendix C: Bequest Motive

TABLE C1: MATCH RATES AND MATCHED PAIR COUNTS

	Matching			Matched pairs in estimation sample			
					Number of children		
Events	Treated households	Matched pairs	Match rate	Pooled	One	Two	Three or more
Death of a child	49,247	23,887	0.485	13,193	1,141	6,378	5,674
Birth of a grandchild	58,971	21,250	0.360	17,179	1,258	9,710	6,211
Child enters disability	188,289	83,211	0.442	62,173	4,803	31,128	26,242

Notes. Table reports the yield of the coarsened exact matching under the specification used throughout, three pre-event and five post-event years. Treated households are all households observed to experience the event under the headline definition of that event. Matched pairs are the treated households assigned a control by the exact match on the variables listed in Panel A of Table C2, and the match rate is their share of treated households. The estimation sample additionally requires both members of a pair to be observed for at least three post-event years; the final three columns split those pairs by the number of children of the treated household.

TABLE C2: COVARIATE BALANCE AT THE MATCHING ANCHOR

	Death of a child		Birth of a grandchild		Child enters disability	
	Treated	Control	Treated	Control	Treated	Control
Panel A. Matching variables						
Birth year	1939.3	1939.3	1941.2	1941.2	1940.5	1940.5
Birth year, partner	1940.2	1940.2	1941.5	1941.5	1941.2	1941.2
Marital status†						
Single	0.475	0.475	0.340	0.340	0.406	0.406
Married or partnered	0.434	0.434	0.619	0.619	0.518	0.518
Divorced	0.006	0.006	0.003	0.003	0.006	0.006
Widowed	0.085	0.085	0.038	0.038	0.070	0.070
Number of children	2.53	2.53	2.41	2.41	2.52	2.52
Number of grandchildren	2.28	2.28	3.05	3.05	3.56	3.56
Wealth decile	5.39	5.39	6.32	6.32	5.35	5.35
Income decile	5.63	5.63	6.57	6.57	5.79	5.79
Age of eldest child	44.3	44.3	40.5	40.5	42.8	42.8
Household age	71.6	71.6	68.1	68.1	70.0	70.0
Panel B. Covariates not matched on						
Mean age of children	41.8	41.4	37.9	38.6	40.1	40.1
Net wealth (€000s)	213.5	209.4	300.8	291.7	199.7	202.5
Gross income (€000s)	42.5	42.9	51.9	50.9	43.5	43.8
Matched pairs	13,193		17,179		62,173	

Notes. Table reports the means of covariates for treated and control households used in the matched difference-in-differences estimation. Panel A includes the matching variables birth year, birth year of the spouse, marital status, number of children and grandchildren, wealth decile and income deciles, average age band of the eldest child, and average household age. Panel B report covariates that are not matched on: mean age of all children of the household, net wealth and gross income. All means are reported at the year treated households are matched to the controls, three years prior to the event time. † Marital status at the anchor year. Single absorbs partnerships dissolved before 2006. Divorced and widowed only count the observed instances since panel entry.

TABLE C3: PANEL ATTRITION AROUND THE EVENT

	Mean years observed				Complete window (%)			
	Pre-event		Post-event		Pre-event		Post-event	
Events	Treated	Control	Treated	Control	Treated	Control	Treated	Control
Death of a child	4.00	4.00	4.69	4.72	99.9	100.0	79.6	81.6
Birth of a grandchild	4.00	4.00	4.89	4.89	100.0	100.0	92.8	92.8
Child enters disability	4.00	4.00	4.76	4.77	99.9	100.0	84.4	85.0

Notes. Table reports the panel attrition around the life events. The first column names the event. The subsequent four columns report the mean number of periods the units are observed in the pre-event and post-event years. The last four columns report the share of households observed fully in the estimation window, before and after the event. Treated and control households are reported separately for comparability of attrition across treated and comparison units.

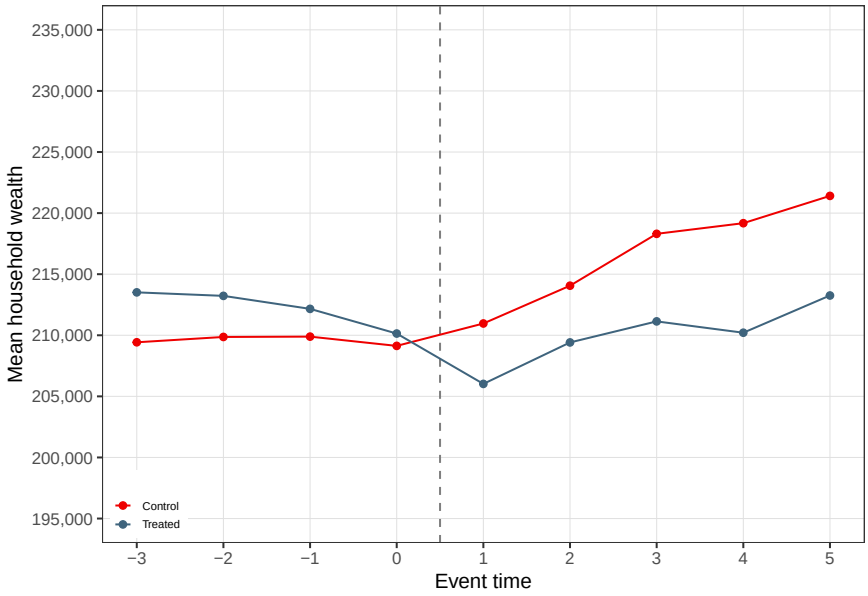
TABLE C4: POST-EVENT EFFECT BY NUMBER OF CHILDREN

	Death of a child Wealth, net-reverse inheritance	**Birth of a grandchild** Wealth	**Child enters disability** Wealth
	(1)	(2)	(3)
One child	$1,279$	140	$-2,444$
	$(4,427)$	$(3,834)$	$(1,620)$
Two children	$-9,327^{***}$	$3,336^{**}$	$-3,199^{***}$
	$(1,790)$	$(1,560)$	(746)
Three or more children	$-10,072^{***}$	$1,915$	$-4,245^{***}$
	$(2,032)$	$(2,037)$	(856)
Mean of outcome	$210,142$	$289,547$	$198,201$
Observations	$225,034$	$298,696$	$1,069,057$
Matched pairs	$13,193$	$17,179$	$62,173$

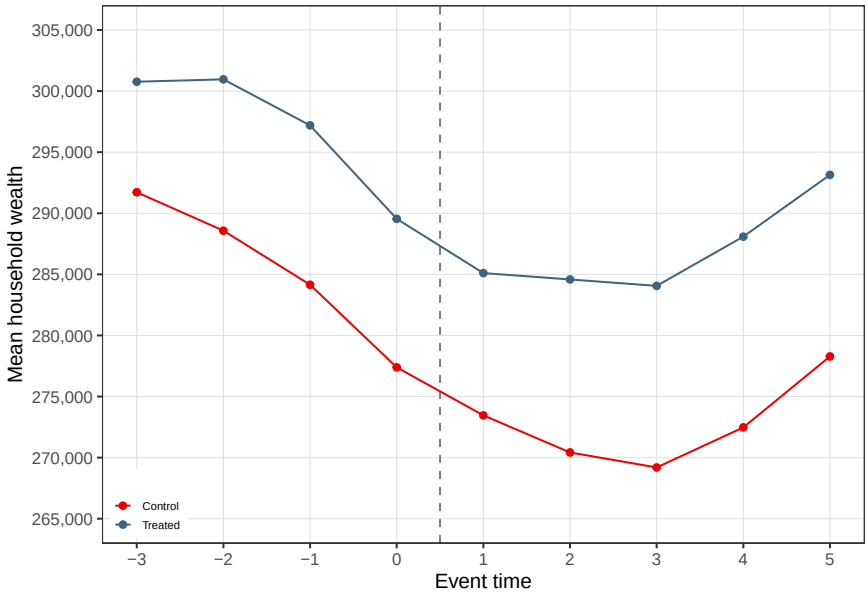
Notes. Reported coefficients are the average of the Sun–Abraham interaction-weighted estimates over event times $k = 1, \ldots, 5$. Standard errors are the average of the corresponding coefficient standard errors, an upper bound on the standard error of the average; significance stars are therefore conservative.

FIGURE C1: WEALTH RAW TIME SERIES AROUND CHANGES IN THE BENEFICIARY SET

(a) Death of a child



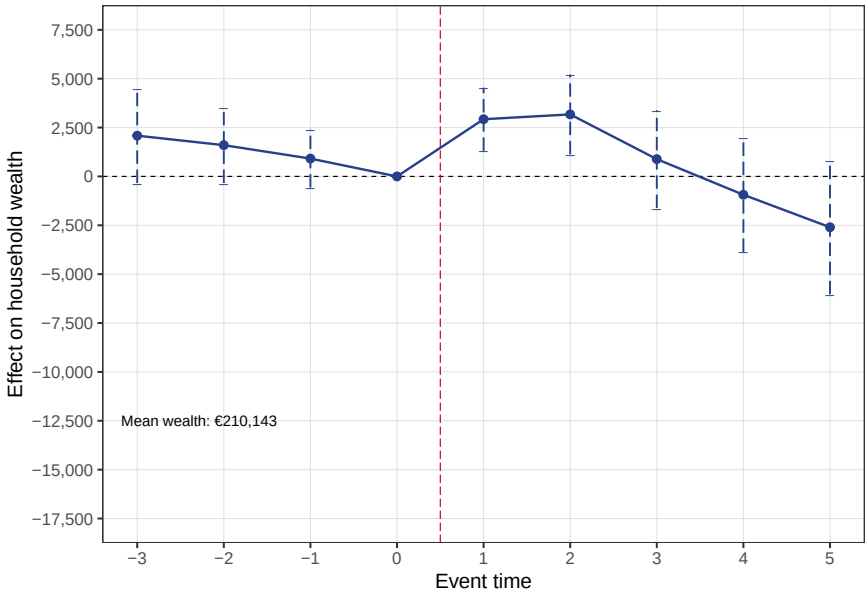
(b) Birth of a grandchild



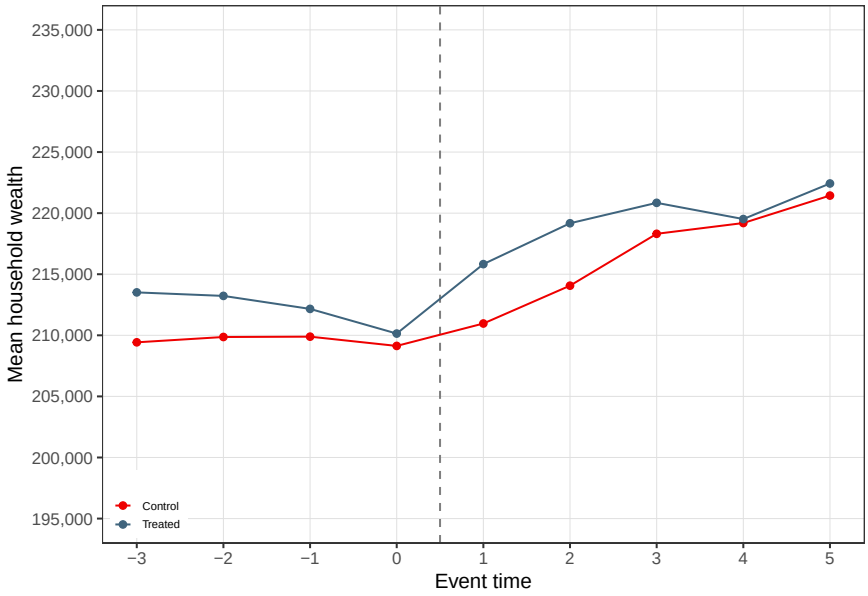
Notes. Mean wealth by event time for the treated households and their matched controls, on the same samples as Figure 5. The horizontal axis is event time, with the event year $k = 0$ normalized to zero. In panel (a) the outcome is household net wealth net of cumulative reverse bequests; in panel (b) it is household net wealth. Confidence intervals are omitted for readability.

FIGURE C2: CHILD DEATH EVENT: ALTERNATIVE OUTCOME

(a) Matched difference-in-differences



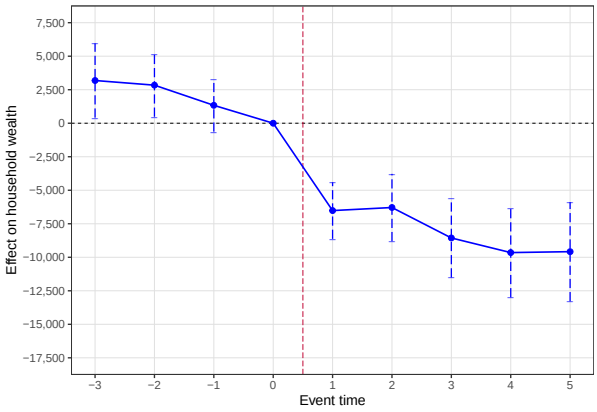
(b) Raw wealth trends



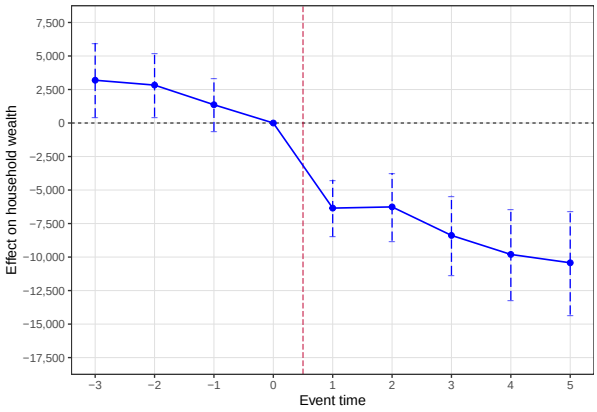
Notes. Figure shows the child death event results from estimating equations 12 and 13 for the Wealth outcome as an alternative to the Figure 5 panel (a) and Figure C1 panel (a), where the outcome is wealth net of reverse inheritance received by the households.

FIGURE C3: WEALTH RESPONSES UNDER THE ALTERNATIVE SPECIFICATIONS

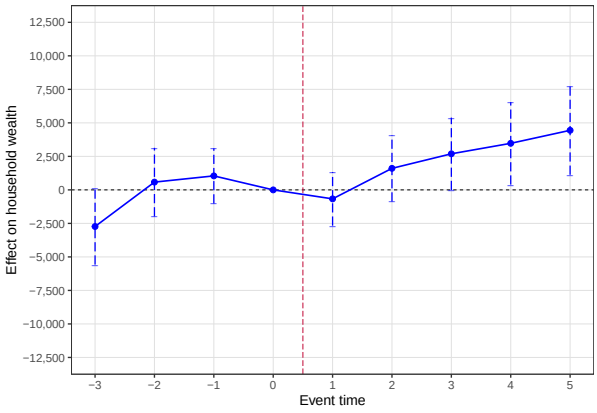
(a) Death of a child
Household and event-time FE



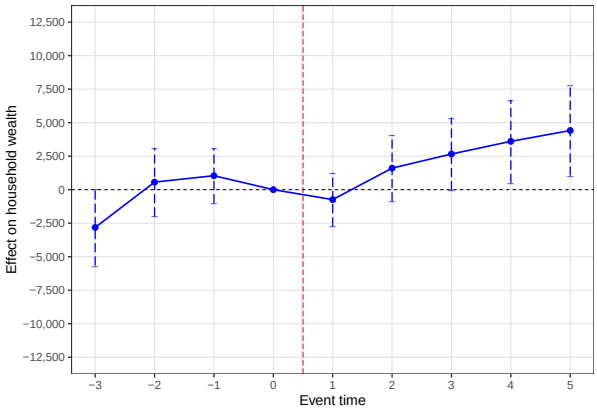
(b) Death of a child
Household and pair × event-time FE



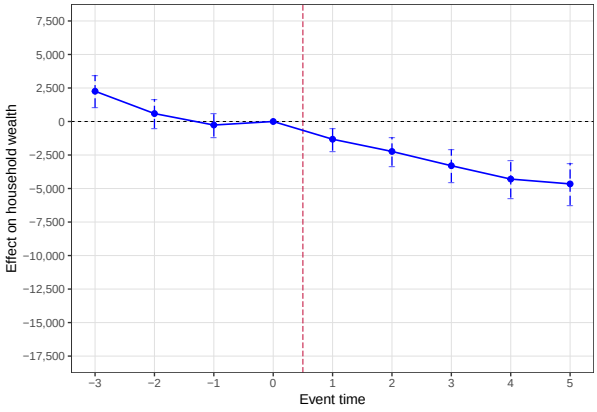
(c) Birth of a grandchild
Household and event-time FE



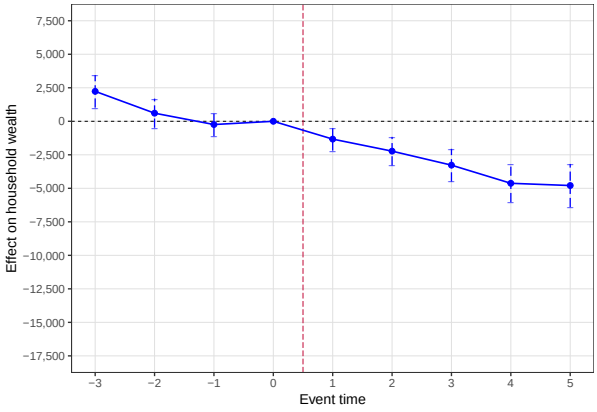
(d) Birth of a grandchild
Household and pair × event-time FE



(e) Child enters disability
Household and event-time FE



(f) Child enters disability
Household and pair × event-time FE



Notes. The figure shows alternative specifications for the three events examined in section 5. Panels (a) and (b) report estimates for child death event; (c) and (d) for grandchild birth event; (e) and (f) for child disability event. For each event, the left plot reports the estimates from a matched difference-in-differences regression with household and event time fixed effects, and the right plot reports the same regression model instead using household and pair times event-time fixed effects.